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DSGE Models in Systematic Trading Strategies

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Abstract

Systematic trading strategies have been researched extensively, capturing tradable patterns in data that are grounded in economic theory and increasingly focus on statistical properties. DSGE models have evolved separately in the field of macroeconomics and provide an effective way of depicting interactions between economic agents with the ability to forecast macroeconomic variables following stochastic shocks. The two areas of study have been explored separately and have had different applications. This has provided an opportunity to test whether there exists value in using DSGE modelling for systematic trading. A basic RBC model is set up and used to generate output forecasts that are compared with realised output data in a novel low-frequency signal creation framework. US equity index assets are traded in this strategy. Results show modest yet healthy performance. Some strategy variants fail to beat the buy-and-hold benchmark, whilst others exceed it and offer insights regarding strategy setup for optimal performance. The exercise proves that DSGE model forecasts and associated equilibrium dynamics offer insights beyond realised data that can be profitably traded in the equities asset class over the long run.

Keywords

DSGE, systematic, trading, strategy, momentum, mean-reversion

(DSGE, systematyczna, strategia, trading, momentum, mean-reversion)

Field of work (Erasmus code)

Economics (14300)

Thematic Classification

Tytuł pracy w języku polskim

Modele DSGE w zastosowaniu w systematycznych strategiach tradingowych

CONTENTS

1. INTRODUCTION

The search for trading strategies is extensive and spans multiple asset classes. Both discretionary and systematic strategies have been researched, tested and implemented by various practitioners and professionals associated with the field of finance. Within the early days of the systematic category, simple investment decisions otherwise performed by analysts and traders were automated (Chan, 2013; Narang, 2013). This gradually became more sophisticated and soon took advantage of increases in computing power and reduced latency to target market phenomena at high frequency (Aldridge, 2013). Such strategies generally attempted to exploit market microstructure nuances often at tick level (Hasbrouck, 2007; O'Hara, 1995). Systematic strategies have become more complex over time, most recently taking advantage of ever-increasing volumes of data and further leaps in computing ability that can help forecast asset price changes, often departing from economic theory in favour of raw statistical accuracy (Gu, Kelly, & Xiu, 2020; López de Prado, 2018).

In a separate world, modern macroeconomics has largely been shaped by a bottom-up style of modelling economic interactions between agents in the economy, characterised as Dynamic Stochastic General Equilibrium (DSGE) modelling (Kydland & Prescott, 1982; Woodford, 2003; Galí, 2015). In one of its basic forms in the Real Business Cycle (RBC) model, households interact with firms. A large number of households seek to maximise their lifetime utility by balancing consumption and leisure. They supply labour and save by investing in capital, which they then rent to firms. Firms maximise profit. They hire labour and rent capital to produce goods based on the available technology and prevailing competitive wages and interest rates (King, Plosser, & Rebelo, 1988). This relationship has its equilibrium and can be affected by stochastic shocks, for example to total factor productivity (TFP) (Kydland & Prescott, 1982). The modelling framework allows for an elegant depiction of the interaction between entities in the economy and to quantify the impact of macroeconomic shocks. Furthermore, DSGE models can be calibrated to suit the characteristics of particular economies (Smets & Wouters, 2003). As is broadly the case in economics, its use case is primarily for inference with the added benefit of forecasting impulse response functions (IRF) of macroeconomic variables under aforementioned shocks (Christiano, Eichenbaum, & Evans, 2005).

On the face of it, it might appear that systematic trading strategies and DSGE models have little overlap, as systematic strategies generally rely on explicit forecasts of asset returns in the hope of generating a signal and opening a position, whilst DSGE models explain the broader economy and are useful for drawing inferences. These models operate at a relatively low frequency in line with the rarer evolution and publishing of economic data which stems from only being measured at such frequency (often quarterly). Nevertheless, the fact that DSGE models offer the ability to forecast macroeconomic variables means that there exist datapoints to which realised values can be compared with the potential of triggering a signal. What remains on the DSGE modelling side is determining the magnitude of shocks, for which estimation techniques can be used, as well as calibrating parameters that resemble a particular economy. Appropriate assets must then be linked to the modelled macroeconomic variables for any signal to hold robustly.

Each of these assumptions and interactions can be explored further. **The intention of this paper is to lay the foundation of a new type of low-frequency systematic trading strategy based on DSGE modelling.** The novelty of the setup requires preparing and running DSGE models, comparing with relevant data and then consolidating into functioning signals for trading. Assets traded in this process should be related to this data economically for the translation of macroeconomic signal to financial return to hold robustly. There exist numerous variations in parameters and assumptions throughout this setup that are worth investigating. This paper explores each of these components and looks to answer the broader question of integration of DSGE modelling in the context of systematic trading strategies. It offers an alternative to established long-term strategies, such as buy-and-hold.

This framework offers further insights as a byproduct: it assesses DSGE model effectiveness by comparing IRF forecasts with realised macroeconomic values in the signal-generation phase and checks the implicit correlation between asset prices and lagged macroeconomic variables via performance of the strategies over time.

It also does not have major barriers to entry from an implementation perspective. Data spans quarterly macroeconomic data from December 1989 to September 2025 and daily asset price series. They do not require specific permissions to be accessed and can, in principal, be substituted with other data. The code is modular and parameters and assumptions can be modified as needed. For example, the rolling portion of the macroeconomic data that is reserved for training the model can be changed, signals can be lagged to the extent required, as well as multiple other potential modifications. The broader infrastructure of the proposed framework defines the Python programming packages and their versions for successfully executing the code and backtesting the strategy variations introduced in this paper. Coupling low dependence on data and being programmed in Python in a specific environment, the framework ensures reproducibility, short implementation time and ease of modification.

2. LITERATURE REVIEW

The development of systematic trading strategies has closely mirrored advances in technology, data availability and quantitative methods. Early systematic approaches largely automated pre-existing discretionary investment rules, allowing traders and institutions to remove emotional biases and improve execution consistency (Chan, 2013; Narang, 2013). These strategies often relied on relatively simple signals derived from technical indicators, statistical arbitrage or trend-following methodologies. Improvements in computing infrastructure and electronic market access gradually enabled higher-frequency implementation, particularly from the late 1990s onwards. This led to the growth of algorithmic and high-frequency trading strategies seeking to exploit temporary inefficiencies in market microstructure, order flow and liquidity provision (Aldridge, 2013; Hasbrouck, 2007; O'Hara, 1995).

The increasing sophistication of systematic trading has subsequently been characterised by a movement away from explicitly theory-driven models and towards empirical optimisation. Modern quantitative investment frameworks frequently rely on large datasets, machine learning techniques and statistical pattern recognition to forecast returns or classify market states (Gu, Kelly, & Xiu, 2020; López de Prado, 2018). Such approaches often prioritise predictive performance over economic interpretability, reflecting a broader trend in finance toward data-centric modelling. This has been facilitated by the expansion of alternative datasets and advances in computational power. However, despite their forecasting capabilities, many machine learning-based strategies remain vulnerable to overfitting, regime dependence and a lack of structural interpretability, particularly during periods of macroeconomic instability.

While advances in machine learning have transformed many areas of quantitative finance, macroeconomic information has continued to play an important role in systematic investment. A broad class of macro-driven trading strategies seeks to exploit relationships between economic conditions and asset prices by incorporating variables such as inflation, economic growth, interest rates, unemployment and monetary policy into the investment process. These strategies are typically implemented through forecasting models, business cycle indicators or econometric relationships rather than structural models of the economy. Consequently, macroeconomic variables are generally treated as predictive inputs whose historical relationships with asset returns are estimated empirically rather than as endogenous outcomes of a fully specified economic system.

Forecasting models represent one of the earliest approaches to incorporating economic information into systematic investment decisions. These strategies typically estimate future values of macroeconomic variables and compare realised outcomes against expectations to identify potential market opportunities. For example, inflation, employment and growth forecasts can be used to anticipate monetary policy decisions and their subsequent impact on asset classes such as equities, bonds and currencies. The importance of macroeconomic surprises has been documented extensively with research showing that unexpected changes in economic indicators can influence asset returns through revisions in expectations regarding future cash flows, discount rates and monetary policy (Chen, Roll, & Ross, 1986; Boyd, Hu, & Jagannathan, 2005). Within systematic strategies, this approach has often been implemented through indicators such as economic surprise indices, where the difference between realised economic data and consensus forecasts is used as a trading signal. Such strategies remain predominantly reduced-form, as the relationship between macroeconomic surprises and asset prices is estimated from historical observations rather than derived from structural economic mechanisms.

A second category of macro-driven strategies focuses on identifying the prevailing state of the business cycle. These approaches recognise that asset returns, volatility and correlations may vary depending on whether the economy is expanding, slowing or entering recession. Consequently, systematic strategies have used business cycle indicators to adjust asset allocation dynamically, increasing exposure to risk assets during favourable economic conditions and reducing exposure during deteriorating conditions. Regime-switching models (particularly those based on Markov processes) have been widely applied to identify changes between economic and market states (Hamilton, 1989; Ang & Timmermann, 2012). Similarly, research on tactical asset allocation has explored the use of leading economic indicators, valuation measures and macroeconomic conditions to adjust portfolio weights over time (Faber, 2007). Although these approaches incorporate economic intuition into systematic decision-making, they generally identify regimes statistically rather than modelling the underlying economic mechanisms generating those regimes. Bhojraj and Swaminathan (2006) provide a relevant example of macroeconomic information being translated into equity return predictability. Their analysis of international equity indices identifies momentum effects linked to macroeconomic conditions with returns displaying persistence over shorter horizons and subsequent reversals over longer periods. This provides evidence that macroeconomic variables and the broader state of an economy can contain information about future equity market performance, whilst also suggesting that the relationship between economic conditions and returns may depend on the investment horizon. The findings are therefore potentially relevant to the present framework, which similarly tests whether macroeconomic information can be translated into trading positions and whether momentum or mean-reversion dynamics are more appropriate. However, unlike the reduced-form relationships examined by Bhojraj and Swaminathan (2006), the present strategy derives its macroeconomic expectations from a structural DSGE model and compares model-implied output against realised output.

A third strand of literature estimates direct statistical relationships between macroeconomic variables and asset prices. These approaches include vector autoregressive (VAR) models, dynamic factor models and other multivariate econometric frameworks that capture interactions between economic variables and financial markets. VAR models have been extensively used to analyse the transmission of macroeconomic shocks into asset prices, including the effects of monetary policy, inflation and economic activity on financial markets (Sims, 1980; Bernanke & Blinder, 1992). Dynamic factor models similarly allow large numbers of macroeconomic indicators to be compressed into a smaller number of latent factors, which can then be used for forecasting and investment applications (Stock & Watson, 2002). More recently, regime-dependent econometric models and machine learning approaches have extended these frameworks by allowing non-linear relationships and changing market conditions to be incorporated into systematic strategies (Ang & Timmermann, 2012; Gu, Kelly, & Xiu, 2020). However, despite increasing sophistication, these approaches remain primarily statistical and do not impose the structural relationships between economic agents that characterise DSGE models.

Whilst macroeconomic information has long been incorporated into investment strategies through empirical forecasting models, structural macroeconomic modelling evolved along a largely separate trajectory. Modern DSGE models emerged following weaknesses of aggregate metrics in macroeconomics. These models sought to explain aggregate economic fluctuations through optimising behaviour of representative agents under rational expectations and stochastic shocks. RBC models demonstrated that relatively simple economies subject to productivity shocks could reproduce several stylised features of business cycles (Kydland & Prescott, 1982; King, Plosser, & Rebelo, 1988). However, the inability of early RBC models to adequately explain nominal rigidities and monetary transmission mechanisms motivated the development of New Keynesian DSGE frameworks.

Subsequent DSGE models incorporated price stickiness, wage rigidities, monopolistic competition and monetary policy rules, substantially improving empirical performance and policy applicability (Woodford, 2003; Galí, 2015). Estimated DSGE models, particularly those of Smets & Wouters (2003) and Smets & Wouters (2007), became central tools within central banks and academic macroeconomics due to their ability to jointly model macroeconomic variables while maintaining structural consistency. Such models offered an internally coherent framework through which macroeconomic shocks could be identified, quantified and propagated throughout the economy. Their widespread use in monetary policy analysis reinforced DSGE modelling as a dominant paradigm in modern macroeconomics.

Despite their prominence in economics, DSGE models have rarely been integrated directly into systematic trading strategies. The majority of macro-driven systematic investment approaches instead rely on reduced-form econometric models, factor investing frameworks or purely statistical forecasting techniques. Traditional macro trading strategies often exploit relationships between interest rates, inflation, growth and asset prices using discretionary interpretation or econometric forecasting rather than structurally derived equilibrium models. Similarly, quantitative asset pricing literature has historically focused on factor models such as the Capital Asset Pricing Model (CAPM) (Sharpe, 1964) and Fama-French extensions (Fama & French, 1993; Fama & French, 2015), which explain expected returns through empirically observed risk premia rather than fully specified macroeconomic equilibria.

There exists, however, a significant body of literature linking macroeconomic conditions to financial market performance. Research has demonstrated that equity returns, bond yields and foreign exchange dynamics respond systematically to inflation surprises, monetary policy shocks and broader business cycle fluctuations (Chen, Roll, & Ross, 1986; Cochrane, 2005). Furthermore, DSGE models have increasingly been used for forecasting macroeconomic variables and analysing policy transmission, particularly through IRFs (Christiano, Eichenbaum, & Evans, 2005). This suggests the possibility that DSGE-implied forecasts or deviations between forecast and realised macroeconomic outcomes may contain exploitable information relevant to asset pricing.

Nevertheless, the literature does not explore the use of DSGE-generated forecasts as direct trading signals within systematic trading frameworks. Existing surveys of quantitative trading indicate that contemporary research has increasingly focused on statistical learning, optimisation techniques, machine learning, reinforcement learning and other data-driven approaches to systematic investment (Jukl & Lansky, 2025). Where DSGE models are applied in finance, they are predominantly used for asset valuation through equilibrium pricing models (Cochrane, 1991), macroprudential stress testing (Alpanda & Zubairy, 2017), scenario analysis of monetary and macroeconomic policy (Woodford, 2003; Smets & Wouters, 2007) or academic asset pricing research (Cochrane, Asset Pricing, 2005), rather than as executable systematic trading strategies. Existing barriers likely stem from several factors. Firstly, DSGE models operate at comparatively low frequency due to the periodicity of macroeconomic data. Secondly, calibration and estimation procedures are computationally intensive and sensitive to modelling assumptions. Thirdly, the translation of macroeconomic variables into tradable asset signals is non-trivial and may vary substantially across economies and market regimes.

This gap in the literature motivates the present study. Whilst systematic trading strategies have increasingly prioritised predictive accuracy without economic structure, DSGE models provide economically interpretable forecasts grounded in equilibrium relationships between agents in the economy. Combining these domains offers the possibility of constructing low-frequency systematic strategies that retain macroeconomic interpretability while producing investment signals upon which to trade. Furthermore, such an approach enables simultaneous evaluation of DSGE model forecasting performance and the strength of relationships between macroeconomic dynamics and financial asset returns.

The contribution of this paper therefore lies not only in proposing a novel systematic trading framework based on DSGE modelling, but also in bridging two largely disconnected strands of literature: structurally grounded macroeconomic modelling and systematic quantitative trading.

Key research questions therefore include:

RQ1: Do DSGE models contain exploitable information for trading?

RQ2: Can DSGE forecasts be appropriately timed with realised data to generate trading signals?

RQ3: Do forecasts lead or follow realised values in generating profitable trades?

RQ4: Do there exist assets that can be profitably traded based on signals related to DSGE modelling?

The main hypothesis is that there exists value in generating DSGE-based forecasts and relating them to realised data for the purpose of trading related assets. DSGE models capture the interaction between agents in an economy and offer a framework for introducing stochastic shocks and producing macroeconomic forecasts. These may provide insights beyond realised data regarding the course of macroeconomic variables, translating into realistic trading opportunities over the long-run. As this framework would involve a novel combination and usage of models, timing model forecasts with realised data for signal generation is a relevant consideration and whose result would likely determine the profitability of the framework. It is also not yet clear whether forecasts of given dates would lead (momentum dynamics) or follow (mean-reversion dynamics) realised values as of the same dates within the trading approach, thus exploring behaviour in that area would be informative. The initial setup would have to trade specific related assets to demonstrate overall feasibility. The aim of this paper is to build the **foundation of a new type of low-frequency systematic trading strategy based on DSGE modelling** that tests the hypothesis. Further details regarding literature review can be found in Appendix B: Extended Literature Review.

3. DATA

The DSGE trading framework in this paper is centred around the US economy and its assets due to availability of data and the prevalence of this country as a benchmark in research. The components of the framework fall into three categories in terms of data: the DSGE model, realised macroeconomic data and assets used in systematic trading strategies.

The DSGE model used in this paper is a basic RBC model defined in 4.1. RBC Model. Whilst the RBC model is largely instantiated with preprepared parameters calibrated to the US economy, it requires a variable component over time which contributes towards signal generation. This is achieved by estimating an autoregressive process of order 1 (AR1) using TFP data. *Nonfarm Business Sector: Labor Productivity (Output per Hour) for All Workers* in the US (OPHNFB) from the Federal Reserve Bank of St Louis (FRED) is a quarterly seasonally adjusted metric used as TFP. Data spans 12/1989-09/2025, amounting to 144 quarters of observations and producing a final signal for use in Q4 2025. Assuming regular full usage of the data as per 4.3. Extensions and Variations, a 10-year initial training period leaves 104 quarters for forecasts and signal generation. The data is visualised in Figure 1.

IRF forecasts of output are one of the results of running the RBC model. These are compared to the realised equivalent variable, which is quarterly seasonally adjusted *Real Gross Domestic Product* in the US (GDPC1) from FRED. Data span is aligned with TFP data and visualised in Figure 1.

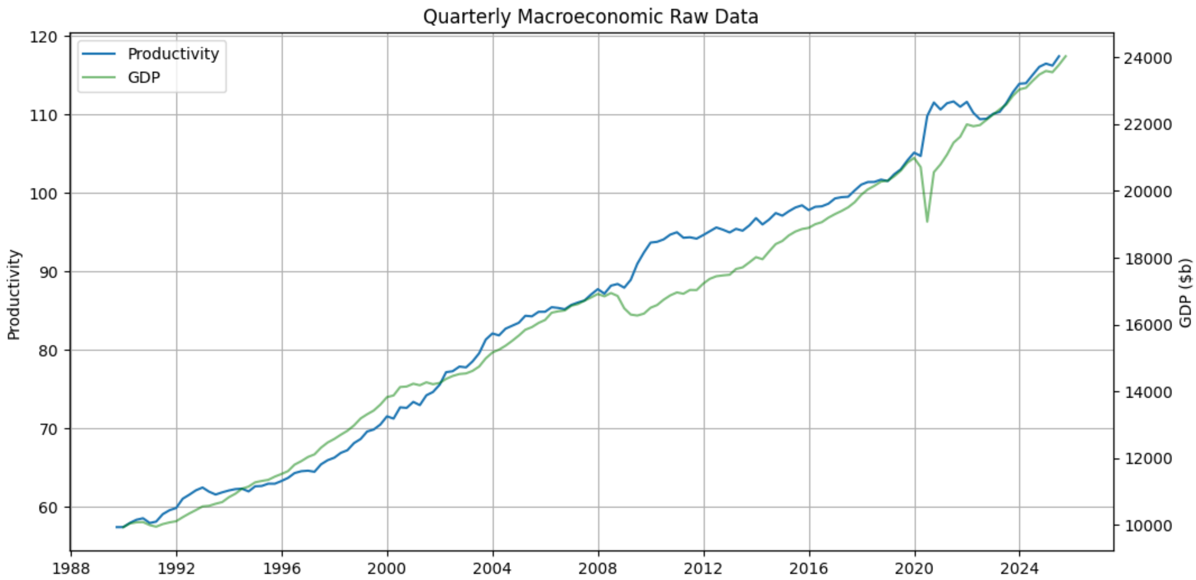

Figure 1: Quarterly Macroeconomic Raw Data

Source: own preparation

Finally, the assets sourced are the daily *S&P 500* index (^GSPC) and the daily *Dow Jones Industrial Average* index (^DJI) from Yahoo Finance, two indices composed of US equities and that are largely reflective of the US economy. In both cases, close prices are used. Data spans 31/12/1999-31/12/2025, amounting to 6540 daily observations. This is equivalent to an average of 252 days per year over that 26-year range, equal to the commonly assumed number of trading days per year. The data is visualised in Figure 2.

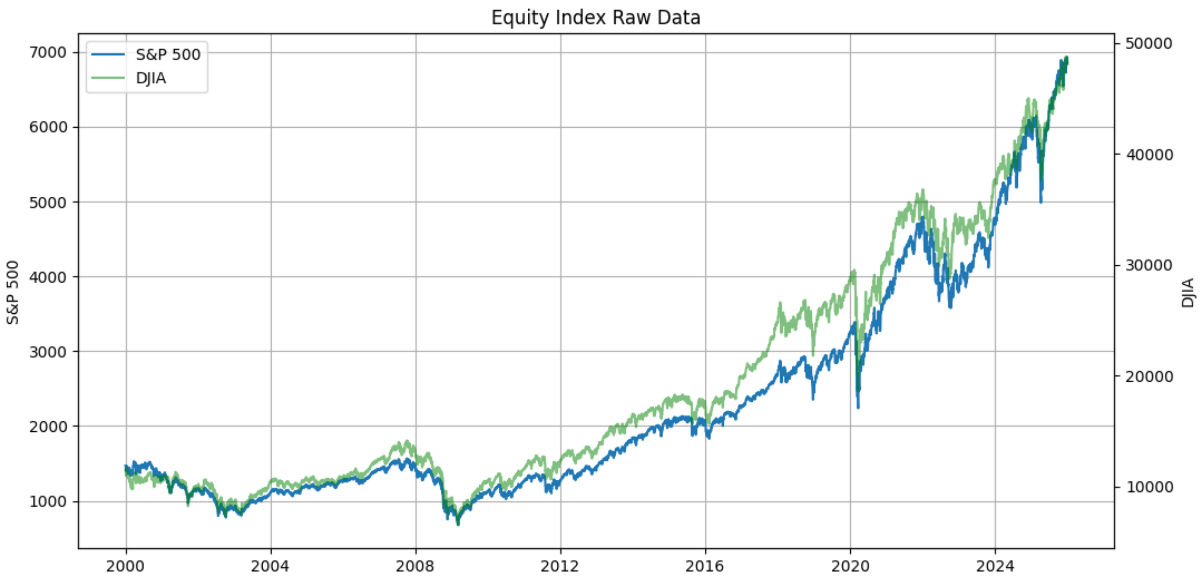

Figure 2: Equity Index Raw Data

Source: own preparation

FRED and Yahoo Finance data are pulled using the Fred class from the fredapi library and the yfinance library in Python, respectively. The limiting factor in terms of time horizons of the data is GDPC1 which is limited to dates after 29/12/1989 in the API, which still ensures a plethora of observations to test multiple strategy variations. All data is pulled once ahead of running any model and is instantiated using an independent data class that spans the maximum possible training and test periods. This prevents spillover between data and its handling, plus avoids having to make multiple slow connections to retrieve data that could be subject to throttling.

There is minimal data cleaning required due to the low frequency and the scrutiny applied by reputable institutions ahead of publishing. Dropping null values does not have an impact due to time series being complete and series indices are formatted to dates. US holidays as per the USFederalHolidayCalendar class in the pandas library are sourced to offset dates to nearest trading days during signal generation.

Overall, data is of sufficient quality for systematic trading strategy application, noting its completeness, healthy observation count, expected frequency and ability to apply through time series operations for strategy backtesting.

3.1. Reproducibility

The design of this framework has kept ease of implementation and reproducibility at its core. The entirety of the framework has been built using Python, including data sourcing for which dedicated APIs are used.

Running the main framework script in a virtual environment with the required libraries installed is sufficient for reproducing the results in this paper. Full instructions are outlined in Appendix A: Framework Setup and in the DSGE strategy repository (Foster & Chlebus, 2026).

4. METHODOLOGY

The approach consists of multiple components described throughout this section. The strategy is fundamentally dependent on the divergence between macroeconomic forecasts and corresponding realised values.

A DSGE model is proposed as a key starting point in emulating the US economy and offers a sophisticated equilibrium-based alternative to simple econometric datapoints. Once this is ready, shocks can be passed through to observe how the system responds. Output is the variable of choice in this process due to its simplicity for comparison with equivalent realised data and broad association with equity markets.

The next stage involves using and transforming GDP data as the realised equivalent series to output. Provided these are timed appropriately, this allows for comparison and subsequent signal creation.

Long and short positions are opened based on such signals: these reflect the view that the DSGE model is either insight-leading in its expectation of an economic state to which realised data is yet to adjust and enabling of a corresponding asset position to close the gap or insight-following in viewing realised data as the direction in which the economy may move towards and justifying an asset position contrary to market sentiment as the market digests an updated economic reality. These positions are further modified based on various parameters in this setup.

Further details regarding methodology in this section can be found in Appendix C: Extended Methods.

4.1. RBC Model

DSGE models describe the economy as a system of optimising agents whose decisions interact through competitive markets. Households maximise expected lifetime utility by choosing consumption, labour supply and saving, whilst firms maximise profits by combining labour and capital to produce output. Prices, wages and interest rates adjust to ensure that markets clear and the economy reaches a general equilibrium. The model evolves dynamically as agents form expectations about future economic conditions and respond to stochastic shocks, such as changes in productivity or monetary policy, allowing the effects of these shocks to be traced through the economy over time (Wickens, 2012).

The RBC model provides the canonical example of a DSGE framework. It describes an economy in which representative households and firms optimise over time in competitive markets with aggregate fluctuations driven primarily by stochastic productivity shocks. Under flexible prices and rational expectations, the interaction between these agents determines the equilibrium evolution of macroeconomic variables such as output, consumption and investment (Kydland i Prescott, 1982; King, Plosser i Rebelo, 1988).

In this model, households maximise expected lifetime utility as per (1):

$$\max_{c_t, n_t} E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, n_t) \tag{1}$$

where β is the discount factor, U is utility, c is consumption, n is labour and t is time.

Firms maximise profit Π as per (2):

$$\max_{k_t, n_t} \Pi_t = y_t - w_t n_t - r_t^k k_{t-1} \tag{2}$$

where y is output, w is wage, r is rental rate on capital, k is capital.

These are subject to the constraints outlined in (3), (4), (5) and (6).

$$c_t + i_t = y_t \tag{3}$$

$$y_t = e^{z_t} k_t^{\alpha} n_t^{1-\alpha} \tag{4}$$

$$k_{t+1} = (1 - \delta) k_t + i_t \tag{5}$$

$$z_{t+1} = \rho z_t + e_{z,t+1} \tag{6}$$

where i is investment, z is TFP, α is elasticity of capital, δ is depreciation of capital, ρ is persistence of the TFP shock.

The setup of the model is available in the dolo Python library from EconForge (Winant, 2019; Winant, 2026). The full system of equilibrium conditions and calibrated parameters can be found in Appendix C: Extended Methods.

The model then becomes suitable for applying stochastic productivity shocks which lead to impulse responses across several macroeconomic variables in the system, for example output and consumption. The magnitude of such shocks and their persistence can be determined via estimation based on productivity data.

For the framework to retain trading applicability, time series data entering the model remains sequential and is used for next-period forecasting. Iterating this approach along a moving window continually fine-tunes model results and applies the latest shock magnitude. This also enables strategy backtesting.

Various training lengths described in 4.3. Extensions and Variations are employed, whereby OPHNFB productivity data is sliced in preparation for the model. This data is converted to natural logarithm form and undergoes HP filtering assuming a quarterly lambda value of 1600 to extract the cyclical component of the time series.

The modified time series z_t is regressed against its lagged values as per the AR1 process in (6) using the OLS estimator. The resulting coefficient is set as the calibrated TFP persistence parameter ρ in the model. For the model to be solved quickly and given the basic RBC model type with small shocks, a local perturbation solution with a first-order Taylor approximation of the policy functions is used. An IRF is then run, assuming the latest TFP shock value $e_{z,t}$ from the residuals in the AR1 estimation.

The goal is to obtain a next-period GDP change forecast. IRF forecast values are multiplicative deviations from the normalised steady state level of 1 (as per Figure 3) and are therefore converted to percentage deviations. The output IRF array is assumed to correspond to GDP. The response at the second horizon period is recorded as the output forecast for this run of the model: the initial response corresponds to the impact period, the response at the first horizon period reflects only the immediate labour and consumption reallocation, whilst the response at the second horizon period acknowledges capital adjustment and reflects the intended impact direction of a productivity shock. It should reflect the timing of quarterly national accounts.

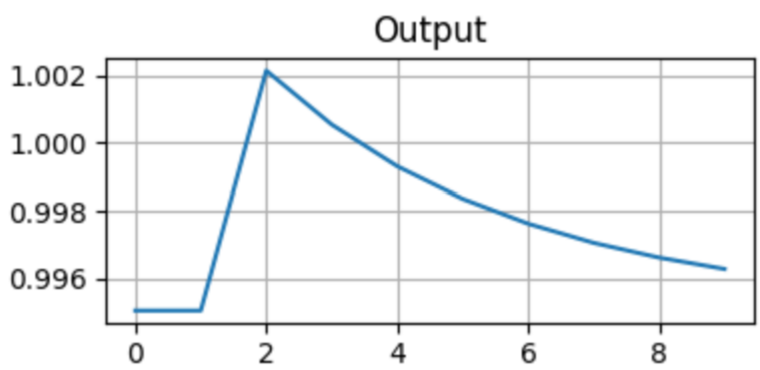

Figure 3: Output IRF assuming rho=0.771 and e=0.0054

Source: own preparation

The process is repeated for data windows shifted quarterly within an overall horizon of the strategy backtest, as illustrated in Figure 4. The result is a time series of next-period output forecasts.

4.2. Signal Generation

The trading strategy depends on both forecast data and realised data. Once these two datapoints are available for a given quarter, their comparison can be used to generate an asset position for the remainder of the quarter. Figure 4 shows the components of the trading strategy and assumed timing. The strategy avoids look-ahead bias by acquiring forecast input data as of a prior quarter, forecasting for a recent quarter, acquiring realised data as of the recent quarter, comparing as of the recent quarter and trading in a subsequent quarter. For example, Q1 forecast input data and Q2 realised data become available soon after the end of Q2; forecasts as of Q2 are produced and compared with Q2 realised data and, following a defined lag after the end of Q2, positions are generated for the remainder of Q3 and until the next signal becomes available. This section explores the details of the process.

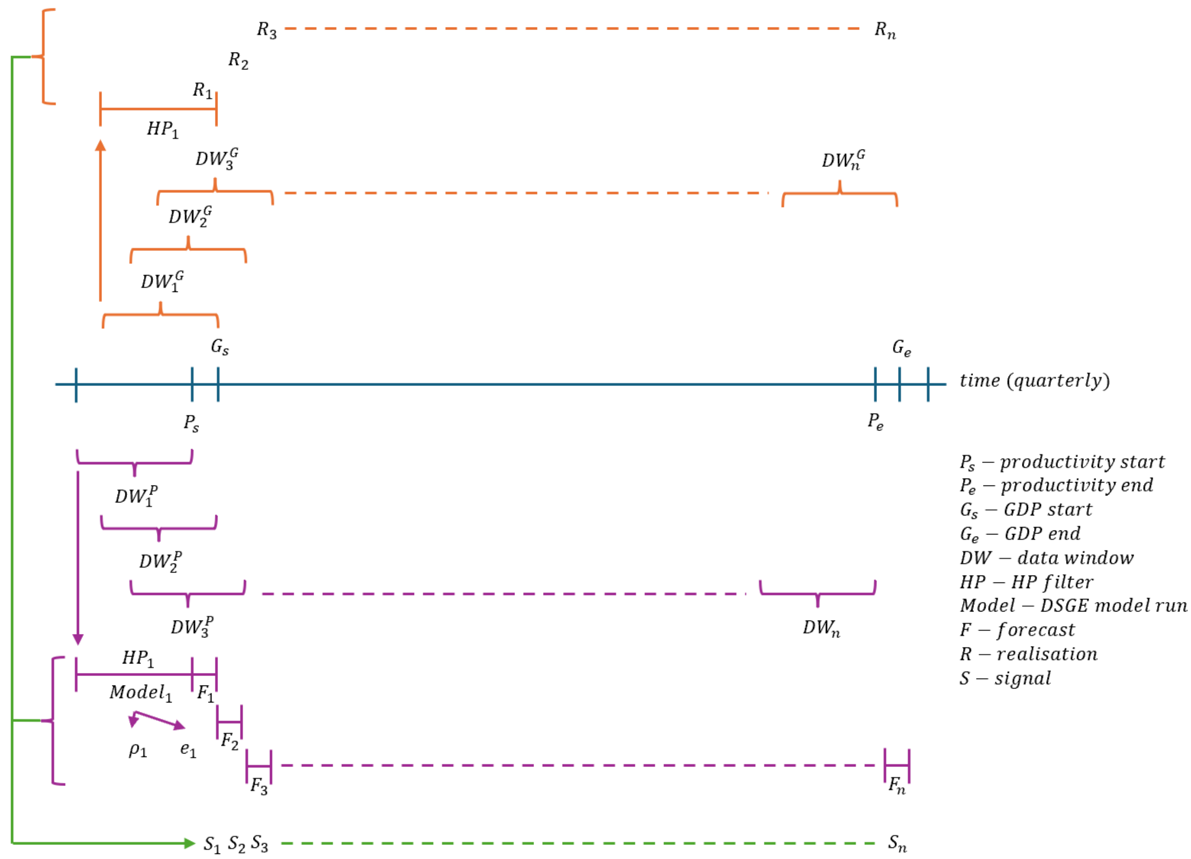

Figure 4: Signal generation illustration

Source: own preparation

Realised values of GDP are needed in addition to output forecasts for subsequent signal generation. GDPC1 data undergoes HP filtering on natural logarithms to ensure a like-for-like comparison with output forecasts. These transformations ensure the resulting series reflect deviations of GDP from trend. The data is initially sliced similarly to OPHNFB data, aligning data lengths but setting the end date one quarter later than OPHNFB corresponding to the date for which the next-period forecast is made. The last element of the array is selected as the realised output value accordingly. The process is repeated for data windows shifted quarterly, forming a time series of realised output values. This is illustrated in Figure 4.

Upon joining output forecasts with realised output, signals can be generated. Two types of signals are implemented: momentum and mean-reversion.

The momentum signal takes a long position when output forecast is greater than realised output, a short position when output forecast is less than realised output and no position otherwise. The intuition in this scenario is that the DSGE model anticipates higher output growth than had been observed, thus a reference asset could be expected to increase in value as output catches up to close the gap.

Conversely, the mean-reversion signal takes the inverse of the momentum position with a view that new information driving divergence in realised output from the forecast implies a pending correction in equity markets and an expectation for them to price in this new information.

Ahead of signal implementation, asset data is prepared for trading. ^GSPC and ^DJI are used in this framework. The two indices are composed of US equities and are largely reflective of the US economy and are therefore natural candidates for the strategy. The daily series of each are restricted to the aforementioned strategy horizon, whereby the last observation ends one quarter after the last signal becomes available. This is due to a period of time required after trading a given signal for the strategy to generate profit and loss (PnL), plus the quarterly frequency of signal changes is in line with the macroeconomic data supplying the model. The series is then converted to percentage returns.

The final modification of the signal series is the crucial delay in its enforcement. In practice, macroeconomic data is released with a lag. Therefore, various lag times are tested in this strategy for the implementation to be feasible and the results to be realistic. For example, a lag of 20 allows 20 business days for GDPC1 data to be released and 20 business days plus one month for OPHNFB data to be released. Signals are further lagged to the nearest trading day if needed for trade execution to be possible.

At this stage signals are joined onto the processed asset data and filled forwards. This effectively sets the position once a quarter and keeps it until the next position is established as shown in Figure 5. The frequency makes it a low-frequency systematic trading strategy with clear room for the assessment of whether the economic phenomena behind the momentum and mean-reversion signals hold.

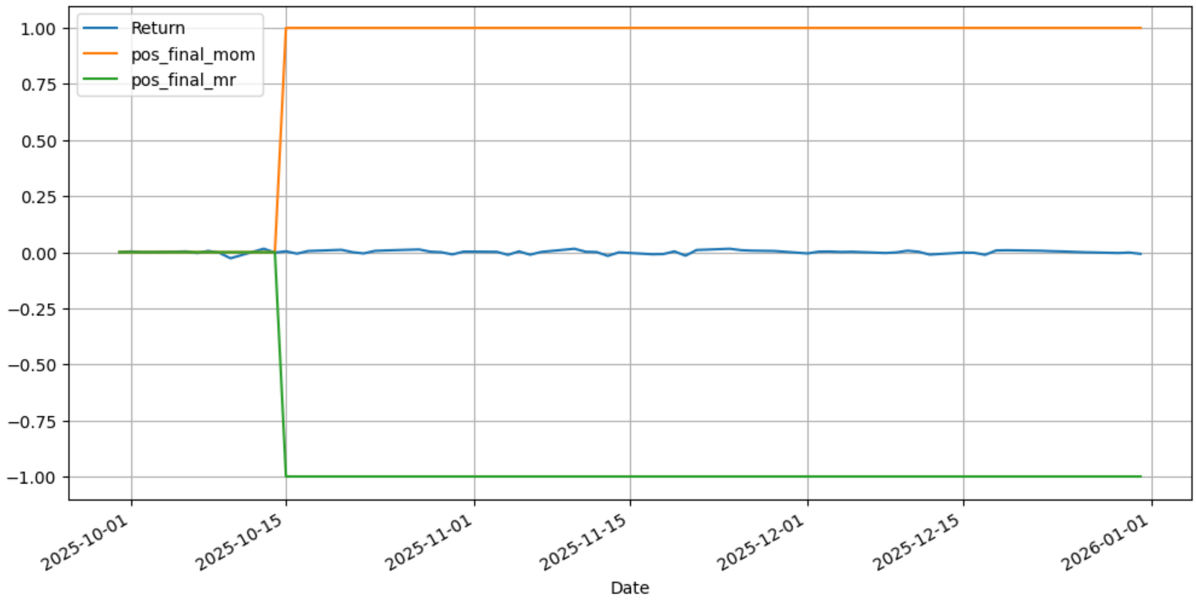

Figure 5: Positioning assuming lag=10 over one quarter

Source: own preparation

4.3. Extensions and Variations

The strategy is extended beyond the aforementioned setup to attempt to improve performance.

Firstly, there exists a doubling down feature. This sets a signal scalar of 2 if two consecutive signals are the same at the quarterly signal generation phase, rewarding signal persistence in each of the momentum and mean-reversion approaches. For example, if output forecasts exceed realised output for two quarters, this doubles the momentum position.

Secondly, leverage can be employed. This is a multiplicative scalar passed down directly to the raw signal. It would be expected to prove useful if baseline results are modestly profitable, few trades occur and profitability is consistent. Results would diverge from doubling down variants if position direction were to change frequently.

Coupling these extensions with variations mentioned throughout this section, there are numerous combinations of parameters in this strategy. These are summarised in Table 1.

Table 1: Strategy parameters

Parameter	**Values**
Model type	Basic RBC
Data length (years for model training)	5, 10, 20
Horizon	12/1999 – 9/2025 12/1999 – 12/2007[1] 12/2007 – 12/2009[2] 12/2009 – 12/2019 12/2019 – 12/2022 12/2022 – 9/2025
Signal lag (days after quarter-end)	10, 20, 30
Leverage	1, 2
Double down	False, True
Asset	S&P 500, Dow Jones Industrial Average
Strategy type	Momentum, mean-reversion

Source: own preparation

Horizon splits are devised as per Table 1, maintaining a balance between the interpretability of multiple versions of results and the ability to capture various parts of the business cycle and note differences in strategy performance over time. Dates rounded to year-ends are used for simplicity. The periods are selected as follows:

- 12/1999 – 12/2007 captures the prosperous build-up towards the Global Financial Crisis (GFC)
- 12/2007 – 12/2009 represents the GFC and resulting economic slowdown
- 12/2009 – 12/2019 covers benign periods and consistently moderate growth periods
- 12/2019 – 12/2022 captures the Covid pandemic and resulting economic crises
- 12/2022 – 9/2025 covers rebound periods and changing geopolitical landscapes

[1] 20-year data length is not available for this horizon, as the earliest observation available is as of 12/1989
[2] 20-year data length is not available for this horizon, as the earliest observation available is as of 12/1989

- 12/1999 – 9/2025 captures the entire available date range for a long-run view of strategy effectiveness

4.4. Costs

Trading costs are considered for a fair and realistic assessment of the performance of the strategy.

$$c_t = \frac{\kappa}{2}|p_t - p_{t-1}| \qquad (7)$$

In line with the literature (Grinold & Kahn, 2000) and as per (7), cost at time t, c_t, is the cost of one full portfolio turnover, κ, scaled by the change in position, p, between time periods. The approach therefore generates a non-zero cost only when a position changes and a trade is required to achieve this. This is further scaled by $\frac{1}{2}$ to avoid double counting trades, for example one unit is charged when changing from a unit long position to a unit short position. The scaling continues to work for larger or levered positions, whose cost grows proportionally to position magnitude.

Throughout the process κ is assumed to be a very low value around 1bp: large-cap US equities have extremely tight spreads and deep order books and their equity indices that are used in this paper further benefit from significant liquidity and brokerage competition (Frazzini, Israel, & Moskowitz, 2014).

Trading costs are expected to have a minimal impact regardless, due to the anticipated infrequent trading activity over the time horizons considered stemming from quarterly signals. This is an advantage of the DSGE strategy that few systematic strategies can afford.

4.5. Testing

The DSGE trading strategy is tested in several ways. Firstly, data inputs are inspected and summarised as in 3. DATA. This includes sense checking observation counts, data availability for the required time periods, time series completeness and a holistic assessment of data usability in the context of systematic trading.

Once the strategies run, PnL and Sharpe Ratio (SR) are calculated. These offer a quantitative method of assessing strategy profitability and performance per level of risk, respectively, and are described in 4.5.1. PnL Calculation and Performance Metrics. These metrics are reported in 5. RESULTS and confirm or reject the momentum and mean-reversion phenomena hypothesised in this paper. They also implicitly assess DSGE model effectiveness via IRF forecasts diverging from realised macroeconomic data and the correlation between asset prices and lagged macroeconomic variables.

Extending the model to variants allows for effective relative testing in 5. RESULTS: as parameters are incrementally modified, impact expectations can be formed and then compared with actual impact of such variants with the goal of confirming the strategy is set up correctly. This is particularly significant for novel frameworks like the one introduced in this paper whereby there are not any suitable direct comparisons for testing. Examples of relative tests include:

- Momentum and mean-reversion strategies should produce directionally opposite performance in a given period
- Leverage and doubling down variants of a momentum strategy that do not affect underlying decisions in signal generation should produce directionally aligned performance as their unlevered counterparts
- Strategies trading S&P 500 should produce similar performance over time as strategies trading DJIA due to the similar price history of both assets, as seen in Figure 2
- Net performance should be very close to gross performance if there are few position changes over the strategy horizon or at most slightly below gross performance

Finally, comparing to a benchmark strategy is an informative test. The benchmark of choice is the buy-and-hold strategy, as described in 4.5.2. Buy-and-Hold Benchmark.

4.5.1. PnL Calculation and Performance Metrics

Gross PnL is calculated daily after incorporating the asset return of the current day and the position of the previous day. Net PnL is calculated by subtracting cost, c_t, from gross PnL every day.

In both cases, PnL reflects the gain or loss of a unit position in a given day. It does not consider the effect of compounding an investment were to have if it were placed at a given point in time and remained in place, reinvesting proceeds with every time period. It is therefore a cumulative arithmetic PnL which scrutinises returns every period and displays lower PnL values than those that would actually be achieved via compounding. For example, a value of 2 indicates a 200% return. This has the additional benefits of comparing signals and strategies more effectively, computing Sharpe ratios and visualising whether a strategy adds value over time.

Sum, mean and standard deviation are calculated across the strategy horizon for both gross PnL and net PnL. In addition, two corresponding variants of the SR are calculated for a risk-adjusted view of the returns of the strategy. Daily SR is converted to annualised SR, assuming 252 trading days per year as in (8) and in line with the trading day count remark in 3. DATA.

$$SR_{annual} = SR_{daily} \times \sqrt{252} \tag{8}$$

4.5.2. Buy-and-Hold Benchmark

The DSGE model-inspired strategy is of low frequency, re-evaluating positions once a quarter. It is expected to have a simple trading profile with few position changes in line with slow-moving economic data. The signals resulting from this framework are applied to equity markets for trading.

These characteristics support the usage of the buy-and-hold strategy as a useful benchmark. The buy-and-hold strategy entails purchasing an asset and holding it for the duration of an investment horizon. The asset is then sold at the end of the investment horizon. Gross PnL is the value of the asset upon sale minus the value of the asset upon acquisition and net PnL further subtracts the costs of entering and existing the position. In practice, the two metrics are expected to be similar, considering the extreme infrequency of trading and long holding periods. The strategy is generally effective over long holding periods whereby the acquired asset may have time to appreciate, for example an equity asset gaining value over time. The buy-and-hold strategy is a well-established benchmark in the literature (Malkiel, 2019).

The benchmark is set up in this framework, ensuring alignment with the DSGE model strategy in terms of asset scope, data used, time horizon and performance metrics.

5. RESULTS

The framework runs seamlessly and produces a wide range of results. This section presents these results in terms of performance and risk and provides analysis of key features and observations.

Gross and net performance are described initially to set the scene of overall strategy performance and explore the impact of trading costs. These are followed by analysis of strategies by variant (horizon, strategy type, data length, signal lag, leverage, doubling down and asset differences) which contain crucial insights regarding strategy setup and performance. The analysis includes but is not limited to an assessment of the varying performance over time, momentum and mean-reversion differences, timing, position scaling and the differences between assets. Presenting the results in this manner modularises them: observations of one characteristic often repeat across remaining components and analysing areas individually adds value towards overall strategy assessment. Finally, comments with respect to the buy-and-hold benchmark serve as additional analysis and put DSGE strategy performance into perspective. Comments with respect to testing and related literature findings are embedded throughout the section.

Ultimately, the results of the DSGE trading strategy and its variants look to address the hypothesis and research questions posed. They provide clear evidence regarding the profitability of such strategies, which of their components are responsible for the performance and further insights regarding modelling and signal timing.

5.1. Gross and Net Performance

PnL and annualised SR are the key metrics used to assess performance of the strategy. Of all horizons, the longest one spanning 12/1999 – 9/2025 is most useful in determining the long-term feasibility of challenging the buy-and-hold portfolio whose position does not change by design and favours long-term holding periods.

Performance varies by each of the hyperparameters with a general skew towards mean-reversion, longer signal lags and leverage for improved results. Mean-reversion strategy variants are the ones that produce positive PnL and SR across the full 12/1999-09/2025 horizon, whilst momentum strategies produce near-mirror losses. The symmetry in PnL profiles and seemingly low magnitudes are expected due to the PnL calculation being a cumulative arithmetic PnL excluding compounding. From a testing perspective, the symmetry additionally demonstrates successful signal implementation. Leverage and doubling down mean-reversion variants boast the largest returns with a maximum of 3.97 net PnL and 3.98 gross PnL for the S&P 500 asset, comfortably exceeding the buy-and-hold strategy that scored 2.03 net PnL and gross PnL. The larger volatility in PnL, however, resulted in lower SR of 0.24, as opposed to 0.4. It should be noted that in both strategy variants leverage is applied to the usual full capital for trading, thus amplifying positions and returns by the leverage scalar, yet also incurring larger trading costs. The most successful unlevered strategy produced PnL of 1.16 and SR of 0.23, lower than the buy-and-hold strategy. This result is not necessarily unexpected and a modest risk-adjusted edge is often achieved despite a real return relationship in the surprise-index literature: an investor would need to correctly trade on ~75% of economic surprises just to match a simple buy-and-hold benchmark (DiNuzzo Wealth Management, 2026). Signal lag is another noticeable feature whose extension generally improves performance, particularly at 30 days lag. Performance metrics are displayed in Table 2 and in full in Appendix D: Extended Results: Strategy Results across Horizons. Each of the components driving strategy variants are explored in detail in 5. RESULTS. Overall results indicate that DSGE models can be used in systematic trading infrastructure and that they contain exploitable information. S&P 500 and DJIA assets can be traded profitably. These may not initially appear as profitable as the buy-and-hold strategy, yet this is challenged in 5.3. Horizon when dissecting component time horizons.

Table 2: Restricted view of S&P 500 results for the full horizon

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.97	3.98	0.24	0.24
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.69	3.7	0.22	0.22
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MR	3.2	3.21	0.19	0.19
RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MR	2.89	2.89	0.17	0.17
BH	N/A	1999-12-31_2025-09-30	N/A	1	No	N/A	2.03	2.03	0.4	0.4
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.16	1.16	0.23	0.23
RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-2.9	-2.89	-0.17	-0.17
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-3.21	-3.21	-0.19	-0.19
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.7	-3.7	-0.22	-0.22
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.98	-3.98	-0.24	-0.24

Source: own preparation

Table 3: Restricted view of DJIA results for the full horizon

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.48	3.48	0.22	0.22
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.32	3.32	0.21	0.21
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MR	3	3.01	0.19	0.19
RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MR	2.81	2.82	0.18	0.18
BH	N/A	1999-12-31_2025-09-30	N/A	1	No	N/A	1.87	1.87	0.39	0.39
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.09	1.09	0.23	0.23
RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-2.83	-2.82	-0.18	-0.18
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-3.02	-3.01	-0.19	-0.19
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.33	-3.32	-0.21	-0.21
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.49	-3.48	-0.22	-0.22

Source: own preparation

It should be noted that whilst both gross and net performance metrics are available, the analysis is devoted to net PnL and net SR. This is due to the realistic assumption of incurring trading costs and due to the alignment with gross metrics in this framework.

Table 2 and Table 3 show the difference is typically negligible with PnL diverging by 0.01 at most between net and gross metric types for both assets. Very few position changes over the horizon explain this phenomenon: there is a total of only 23 position changes for the unlevered 10 year-calibrated full horizon strategy variants, each of which is inexpensive. This equates to a charge applied only 0.35% of the time out of 6540 daily observations throughout the horizon. Low turnover is generally acknowledged as a cost advantage (Grinold & Kahn, 2000; Frazzini, Israel, & Moskowitz, 2014). The effect is visualised in Figure 6. Equivalently, the mean holding period in these strategy variants is 3.15 quarters.

In fact, all full horizon strategy variants ranged 23-37 trades as per Table 4. Therefore, the focus throughout the remainder of 5. RESULTS is on the more realistic net performance.

Table 4: Position changes in 12/1999-09/2025 strategies

Strategy	Position Changes Count	Position Changes Percentage (%)
RBC 5y 1999-12-31_2025-09-30 DD (any lag, any leverage)	37	0.57
RBC 10y 1999-12-31_2025-09-30 DD (any lag, any leverage)	32	0.49
RBC 5y 1999-12-31_2025-09-30 (any lag, any leverage)	25	0.38
RBC 10y 1999-12-31_2025-09-30 (any lag, any leverage)	23	0.35

Source: own preparation

Buy-and-hold strategy results assume a similar trading cost profile, as the long position remains once initiated. Therefore, there is no need to distinguish between gross and net performance for the benchmark strategy.

The low trading frequency is a key advantage of the DSGE strategy framework and it reinforces the view that the framework is capable of sustaining a net profit over the long-run.

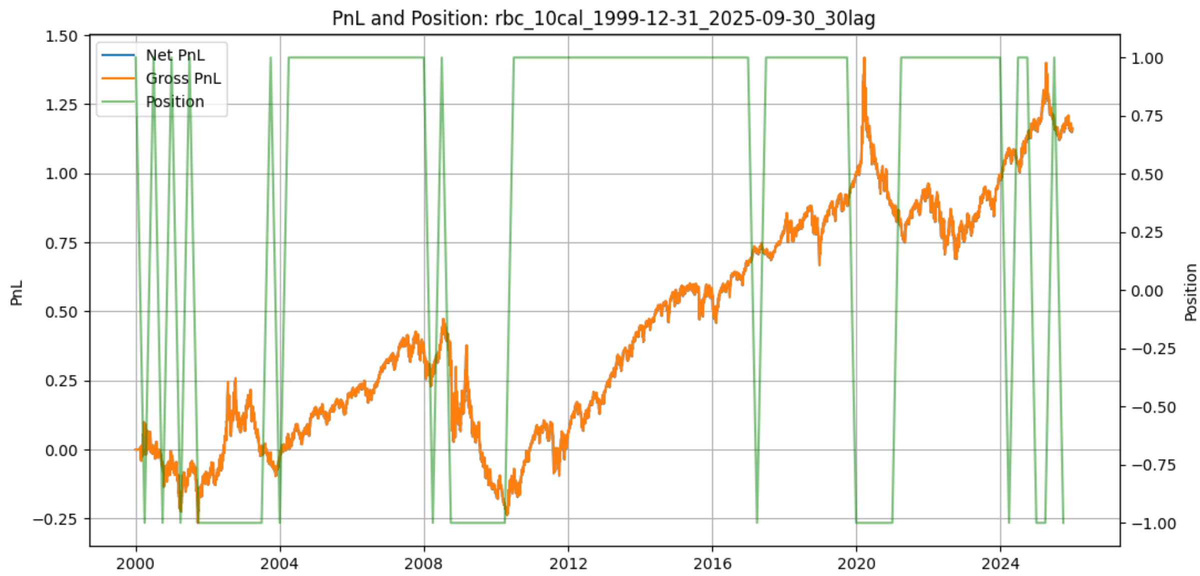

Figure 6: PnL and positions for RBC 10y 1999-12-31_2025-09-30 30b lag MR strategy

Source: own preparation

5.2. Strategy Type

Mean-reversion strategies perform better than momentum strategies in the DSGE trading framework. Table 2, Table 3 and Appendix D: Extended Results: Strategy Results across Horizons show the dominance of mean-reversion in terms of PnL and SR, regardless of training data length, signal lag and other attributes. This suggests that positioning in line with how output beats or misses expectations is key in attempting to capture alpha in equity markets: the more output exceeds forecast values, the more it could indicate the economy is outperforming and that equities are undervalued and worth taking a long position in; equally the lower output turns out to be relative to the forecast, the more it could suggest the economy is underperforming and a stock market correction is due. Realised data may assume more information about the state of the economy and react to additional short-term shocks (Baltussen & Soebhag, 2022).

The association between output performance and subsequent equity market moves is evident through the long-term consistency of the strategy, yet not as strong as other potentially shorter-term variables, as net SRs are below buy-and-hold net SRs for several unlevered strategy variants. This is illustrated for variants of the full horizon strategy in Figure 7.

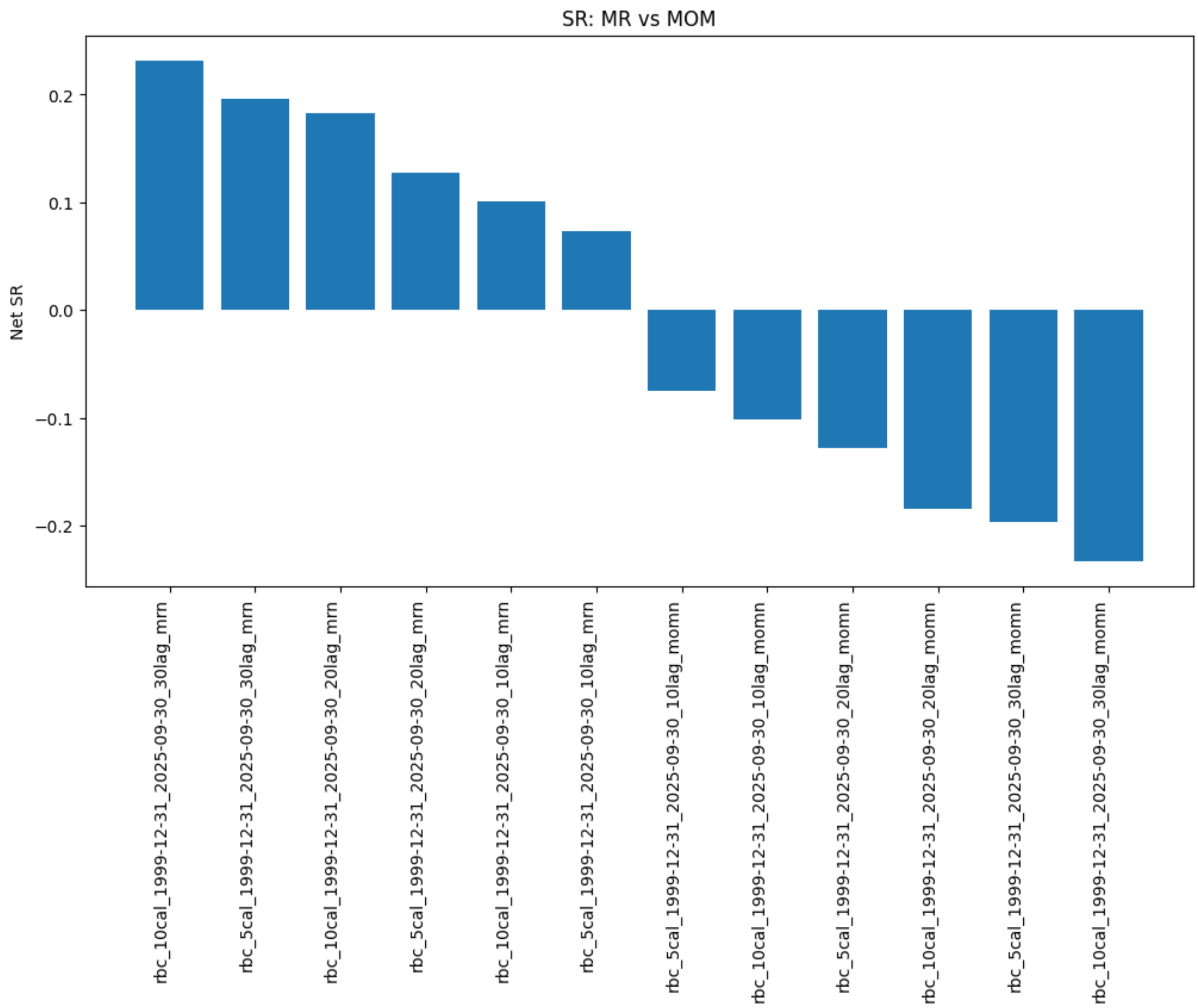

Figure 7: MR and MOM Sharpe ratios[3]

Source: own preparation

Furthermore, mean-reversion strategies are unprofitable during some periods of time whilst momentum strategies generate positive PnL, as discovered in 5.3. Horizon. The association may therefore be the inverse during such periods, particularly during periods of increased volatility and financial crises. Over GFC and Covid periods, it can be seen that the forecast drives the direction for a profit-generating signal: as the economy encounters negative productivity shocks and resulting output forecasts are lower than realised output, generally taking a short position with a view that corresponding equity markets have more to lose earns

[3] Strategies are labelled as per Appendix F: Strategy Labelling Convention

positive PnL. More importantly, following forecasts after initial market shocks subside and into the recovery phase is key for profitability, as equity markets price in a series of anticipated interest rate cuts and depart from the actual state of the economy soon after positive forecasts beat realised output. Similar observations are found in the literature whereby there exists momentum in country/macro-linked equity indices in the near-term with reversals emerging over subsequent years (Bhojraj & Swaminathan, 2006).

The overall mechanism appears to favour mean-reversion towards realised output in the application of a trading signal to assets linked to the economy, which in this case are two major US equity indices. This reverses during volatile and troubling economic periods whereby pursuing the direction of the forecast instead is beneficial. This therefore answers the research question on forecasts leading or following realised values in generating profitable trades.

A separate observation is that the consistency with which momentum and mean-reversion variants all either generate positive PnL or negative PnL demonstrates the appropriateness of the strategy setup. The opposite directions of momentum and mean-reversion variants under the same remaining parameters indicate signals are being generated as intended and driving contrasting long and short positions.

5.3. Horizon

Whilst full horizon variants of the strategy are most useful in determining long-term feasibility of the DSGE trading framework, it is worth exploring how the strategies fared throughout this period, spanning various economic conditions and asset fluctuations.

Among the strategies that did not employ leverage, the most successful one for the S&P500 asset across the full horizon was the mean-reversion one with the RBC model, AR functions trained over 10 years and a signal lag of 30 business days. It produced net PnL of 1.16 and net SR of 0.23. Its comparison with the strategies parametrised the same way except for horizon is shown in Table 5 and Figure 8. The strategy peaks at 0.81 net SR in 12/2009-12/2019 and experiences a trough of -0.67 net SR in 12/2007-12/2009.

Table 5: Strategy comparison across horizons for S&P 500

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Net SR
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.27	0.19
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.5	-0.67
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.4	0.81

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Net SR
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.25
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.76
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.16	0.23

Source: own preparation

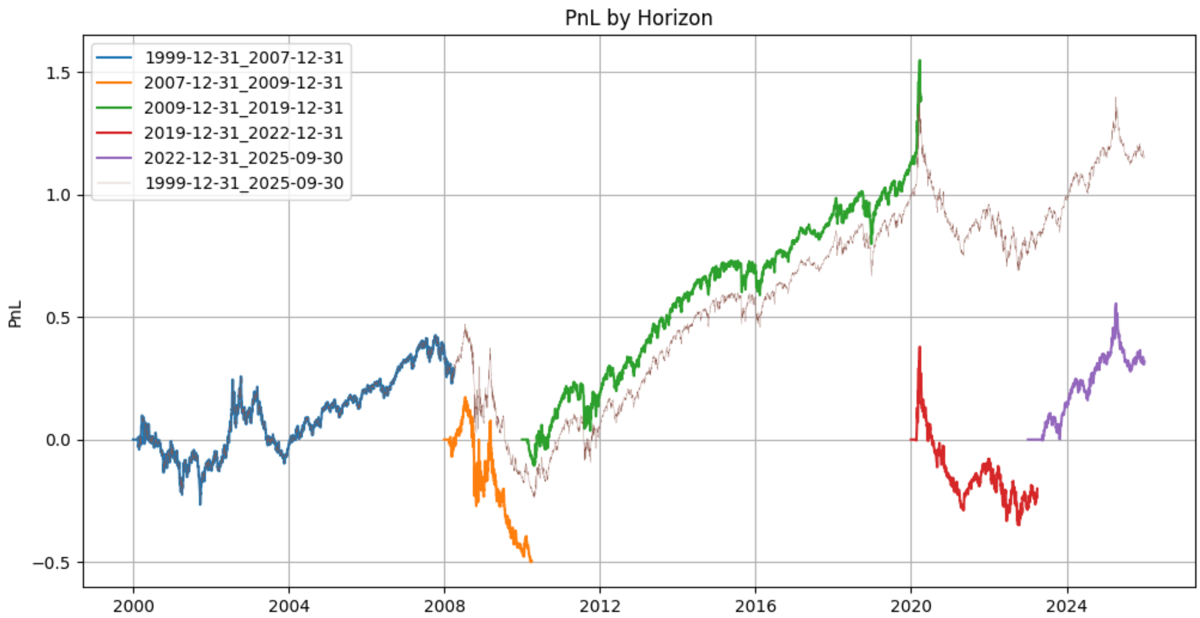

Figure 8: Strategy comparison across horizons for S&P 500

Source: own preparation

Despite overall Sharpe ratios not indicating market-beating risk-adjusted performance, it can be observed that the strategies generally perform healthily and are consistent. In the mean-reversion (MR) scenario, a short position is adopted when output forecast exceeds realised output and a long position otherwise. Throughout most of the bull markets of the mid-2000s, 2010s and mid-2020s the signal captures output beating forecast expectations, suggesting better-than-expected economic growth and corresponding stock market rallies. It profitably holds long positions for most of those time periods accordingly. Impressively, it shorts the S&P 500 several times in the first half of the 2000s through a series of occasions when realised output does not meet forecast expectations and a bear market ensues. Conversely, it loses money in the early 2020s when it intuitively goes short into an economic downturn driven by the Covid pandemic. The stock market had quickly reacted positively following interest rate decreases and a technology firm-driven rally, thus it can be argued there existed a greater decoupling of financial markets from the real economy. The strategy variant running throughout 2008-2009 also loses ground, although this is in line with the buy-and-hold strategy.

The strategies are somewhat stable in their respective horizons and are expected to be as such due to the slow-moving nature of economic data and especially its forecasts. Moreover, the simplicity of the basic RBC model used throughout these variations ensures consistency and is well adapted to the frequency of the data, preventing overtrading. This is particularly useful in extended bull runs, as observed in 2010s where the strategy is profitable over the long-run and remains committed to positions. The additional benefit is the ability to identify several-quarter periods of acyclical changes in output, as in the mid-2000s, to which equities appear to be correlated and from which the signal is able to generate positive PnL whilst the buy-and-hold strategy does not.

For the framework to be profitable throughout all periods, it would require selecting the momentum strategy variants around the times of the GFC and Covid. As per Appendix D: Extended Results: Strategy Results across Horizons, these returned a respectable 0.54 net PnL (0.72 net SR) and 0.51 net PnL (0.32 net SR), respectively. This suggests that the forecast becomes the more valuable source of information during a downturn and, with it anticipating output lower than the output observed following negative productivity shocks, it correctly signals that equity markets are due to decrease further. This also addresses one of the research questions: forecasts can both lead and follow realised values in generating profitable trades and this feature changes depending on the time horizon. In contrast, the literature finds that conventional momentum strategies are prone to sharp reversals in 'panic' states following market declines and elevated volatility (Daniel & Moskowitz, 2016). This is due to the signal in this framework capturing macroeconomic forecast-realisation gaps rather than standard price momentum.

Setting momentum variants in place of mean-reversion ones for the aforementioned horizons produces lucrative results whereby net PnL reaches 2.68, at which point it comfortably beats the buy-and-hold strategy. The relevance of this is that it offers the potential to beat the well established strategy in a like-for-like setting: there is no leverage employed, the strategy horizon remains long-term and there are minimal costs involved in trading. The only requirement for this to function is the existence of a trigger of regime change. This may not necessarily be trivial as a general exercise as has been documented in the literature (Ang & Timmermann, 2012; Hamilton, 1989), yet there are factors that improve the chances in a future application to this framework. Firstly, the time between successive position changes amounts to a full quarter which leaves ample time to initiate a change in strategy type based on financial or economic data, for example yield curve inversions, poor Purchasing Managers' Index (PMI) readings or sudden geopolitical crises. Secondly, the systematic trading strategy is long-term by design, thus as impending crises or regime changes become apparent to a wide audience, strategy type can be switched to reap the rewards for the several quarters for which positions remain fixed on average. As a result of the superior returns of this hybrid approach (yet noting the constraint to identify regime changes), it may be considered a realistic alternative to buy-and-hold and further makes the case for the hypothesis that there exists value in generating DSGE-based systematic trading frameworks. Figure 9 visualises the impact of the hybrid approach.

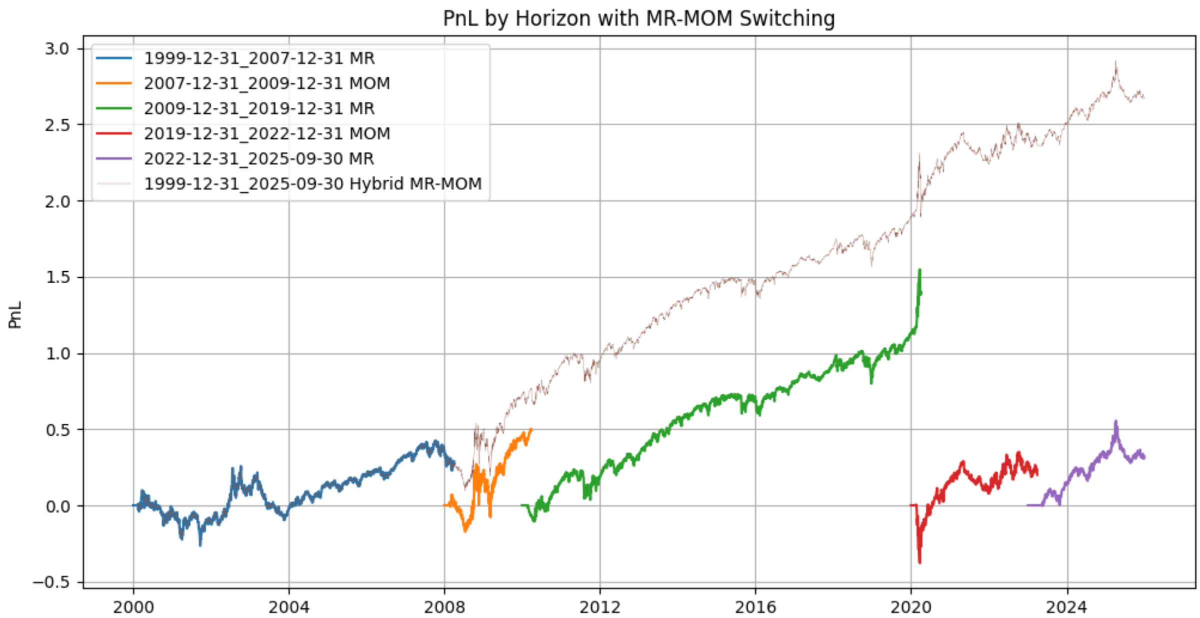

Figure 9: Strategy comparison across horizons for S&P 500 with strategy type switching

Source: own preparation

The results across strategy horizons of the Dow Jones Industrial Average (DJIA) asset are aligned with those of the S&P 500, however net PnL and net SR are lower for this asset. This is captured in Table 6 and Figure 10. S&P 500 has slightly larger net SR in absolute value than DJIA for all horizons except 12/2022-09/2025. The impact of greater PnL outweighs its greater standard deviation, except for the final horizon in which the S&P 500 strategy experiences greater volatility and a lower cumulative PnL. The finding of overall profitability and horizon differences holds for the DJIA asset.

Table 6: Strategy comparison across horizons for Dow Jones Industrial Average

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Net SR
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.18	0.13
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.36	-0.54
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.27	0.74
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.19	-0.23
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.36	0.95
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.09	0.23

Source: own preparation

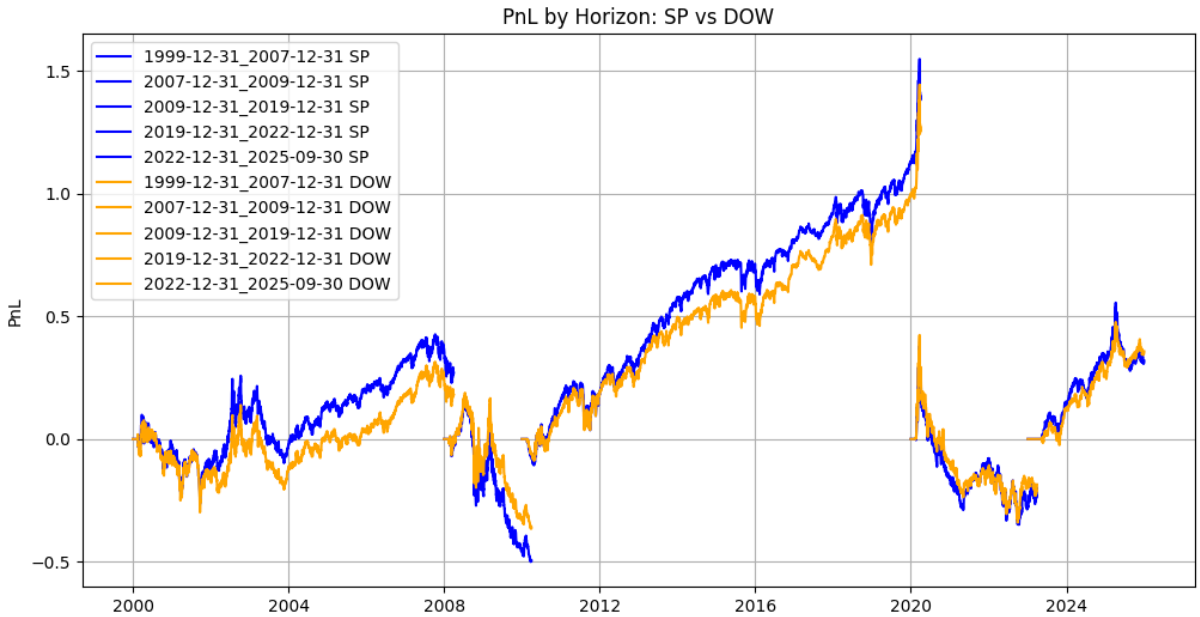

Figure 10: Strategy comparison across horizons for S&P 500 and Dow Jones Industrial Average

Source: own preparation

5.4. Data Length

Three training data lengths are employed throughout the framework. In line with the long-term nature of the framework, these lengths span multiple years for AR equations to produce feasible results and meaningful productivity persistence parameters for the model.

The 5y strategy was less reliable, producing a larger net SR for the 12/1999-12/2007 period, yet performing noticeably worse in 12/2022-09/2025 as shown in Table 7. Its erratic nature stems from a shorter time period to form robust persistence parameters and the susceptibility to being swayed by several volatile quarters of TFP data. The HP filter also likely struggles to split cyclical and trend components for data this short.

The 20y strategy fares slightly worse than the 10y strategy across variants, plus does not produce results up to 12/2009 observations due to the insufficient data availability.

10 years are most effective in increasing net PnL and net SR across horizon and signal lag variants for unlevered mean-reversion strategies. This demonstrates that the DSGE trading strategy can be profitably traded under appropriate modelling choices. Results for more signal lags can be found in Appendix E: Extended Results: Strategy Results across Data Lengths.

Table 7: Unlevered S&P500 Data Length Results for 30b signal lag for all horizons

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Net SR
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.16	0.23
RBC	5y	1999-12-31_2025-09-30	30b	1	No	MR	0.98	0.2
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MR	0.56	0.39
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.27	0.19
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.5	-0.67
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MR	-0.56	-0.76
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.4	0.81
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MR	1.25	0.72
RBC	5y	2009-12-31_2019-12-31	30b	1	No	MR	1.18	0.68
RBC	5y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.25
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.25
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.25
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.76
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.76
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MR	0.18	0.43

Source: own preparation

5.5. Signal Lag

Three signals lags are run in the DSGE framework. These delay the relation of output forecasts to realised output values before triggering the opening of a position.

As with the 10y data length conclusion, the 30b signal lag consistently outperforms other variants for unlevered mean-reversion strategies: it typically ranks first in terms of net SR and, when second, it produces results close to that of the winning variant, as portrayed in Table 8. A general rule of thumb appears to be that the longer the lag, the more profitable the resulting strategy. This implies that the factors underlying divergence of realised output from forecasts begin to take effect in equity markets at least a month after the reference date and continue to translate to corresponding valuations throughout the rest of the quarter for mean-reversion variants. This is aligned with literature findings whereby price drift persists for at least 60 trading days after the announcement of an earnings surprise and which gives empirical precedent for information from a scheduled fundamental release taking weeks to be fully priced (Bernard & Thomas, 1989).

A longer lag length is also more practical in terms of signal generation: 30 business days allow more time for macroeconomic data providers to publish TFP and GDP data and for the data to be processed for signal generation.

Table 8: Unlevered S&P500 MR signal lag results for 10y data length for all horizons

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Net SR
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.16	0.23
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MR	0.92	0.18
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MR	0.51	0.1
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.27	0.19
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MR	0.16	0.11
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MR	-0.16	-0.11
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MR	-0.43	-0.58
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.5	-0.67
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MR	-0.54	-0.72
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MR	1.42	0.82
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.4	0.81
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MR	1.3	0.75
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.25
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.32
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.35
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.76
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MR	0.27	0.64
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MR	0.16	0.38

Source: own preparation

The reverse is true for momentum strategies in Table 9: 10b appears to produce higher net SR. This implies that forecast supremacy relative to realised output takes less time for equities to price in for momentum variants. In other words, the sooner a position is taken following the realisation that output is not at the level its forecast suggests it should be, the more likely it is that a corresponding gain can be made in the associated asset. However, this phenomenon is not as clear as with the 30b signal lags in mean-reversion whereby the 10b lag does not end victorious in two out of six momentum horizons.

Table 9: Unlevered S&P500 MOM signal lag results for 10y data length for all horizons

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Net SR
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MOM	-0.51	-0.1
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MOM	-0.92	-0.18
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MOM	-1.17	-0.23
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MOM	0.16	0.11
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MOM	-0.16	-0.11
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MOM	-0.27	-0.19
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MOM	0.54	0.72
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MOM	0.5	0.67
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MOM	0.43	0.58
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.3	-0.75
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.4	-0.81
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.42	-0.82
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MOM	0.29	0.35
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MOM	0.25	0.32
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MOM	0.2	0.25
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MOM	-0.16	-0.38
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.27	-0.64
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.32	-0.76

Source: own preparation

Overall, these observations and practical ease of implementation point to 30b being the preferred signal lag. This concludes the research question on appropriate timing of DSGE forecasts and realised data to successfully generate trading signals.

5.6. Leverage and Doubling Down

Directly applying leverage as a scalar and doubling down on persistent positions collectively represent the concept of leverage. Trading directions in these strategy variants are aligned with their unlevered counterparts, yet the magnitudes of resulting positions diverge. Table 2 and Figure 11 show that they can have positive effects on net PnL.

The approach to signal generation and position setting in this framework is conducive to applying leverage. Positions change infrequently and, provided the right mean-reversion strategy type is selected and the portfolio is established in a benign regime, the portfolio is well prepared to earn excess returns and cover higher trading costs. Furthermore, the assets chosen for trading in this framework consist of a broad range of equities whose long-term cumulative value generally increases. However, the benefit is path-dependent and may not materialise under high volatility.

Doubling down is very similar to 2x leverage in the DSGE framework. This is due to long periods of time during which positioning remains unchanged, hence doubling down sets in frequently.

Combining the impact of both scaling approaches results in an effective near 4x leverage and final net PnL of 3.97 in the 10 year-calibrated 30 day signal-lagged full horizon mean-reversion strategy for the S&P 500 asset. Net SR remains at 0.24 due to the increased volatility of PnL and is expected in the absence of variance-conditioning dynamic leverage techniques (Daniel & Moskowitz, 2016).

The impact reinforces the view that there exist assets that can be profitably traded based on signals related to DSGE modelling and reaffirms the hypothesis that there is value in pursuing this framework.

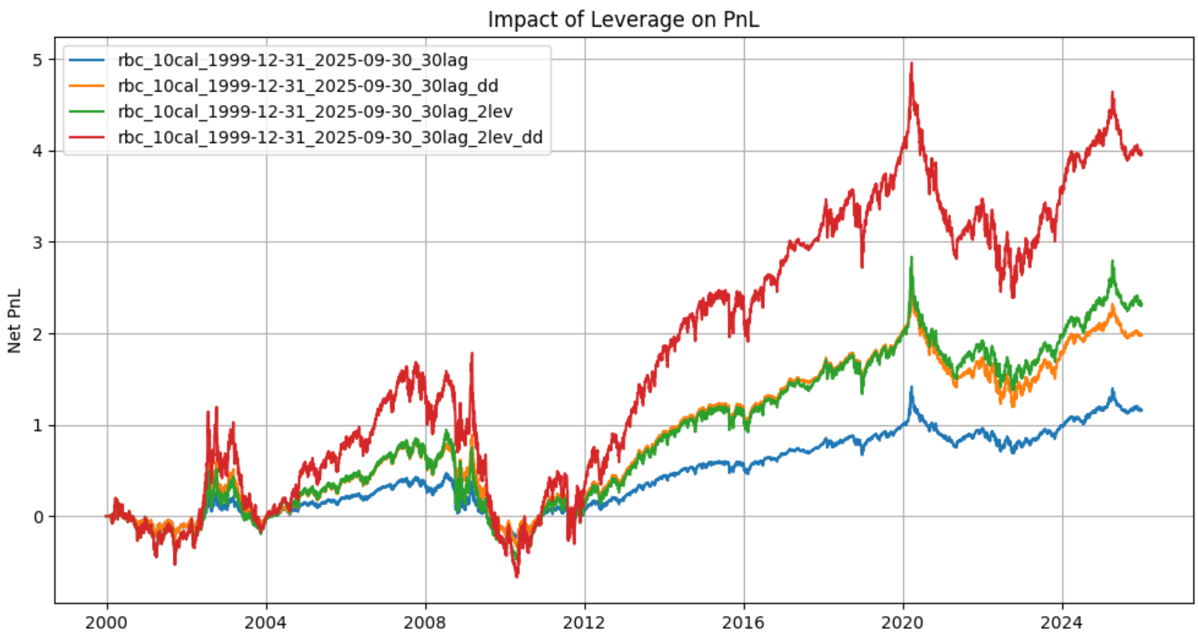

Figure 11: PnL with doubling down and 2x leverage of RBC 10y 1999-12-31_2025-09-30 30b lag strategy

Source: own preparation

5.7. Buy-and-Hold Benchmark and Asset Differences

The buy-and-hold strategy prints a net PnL of 2.03 and net SR of 0.4. Being long one unit in an equity asset, it performs well in bull markets and poorly in bear markets. Its infrequent trading, long-term investment horizon and applicability to the equities asset class render it an acceptable benchmark for the DSGE trading strategy (Malkiel, 2019).

Whilst delivering on the objective of systematically earning a net profit, the best unlevered DSGE strategy performs slightly worse than the buy-and-hold strategy over the full horizon. Aforementioned nuances described throughout 5. RESULTS improve performance, for example adding leverage or introducing logic that switches between mean-reversion and momentum depending on the regime. Net PnL progression over time is illustrated in Figure 12.

Results of the DJIA asset are comparable to those of the S&P 500 and conclusions are aligned between them. There are not any meaningful differences observed across strategy variants. Relevant analysis can be found in 5.3. Horizon and 5.1. Gross and Net Performance.

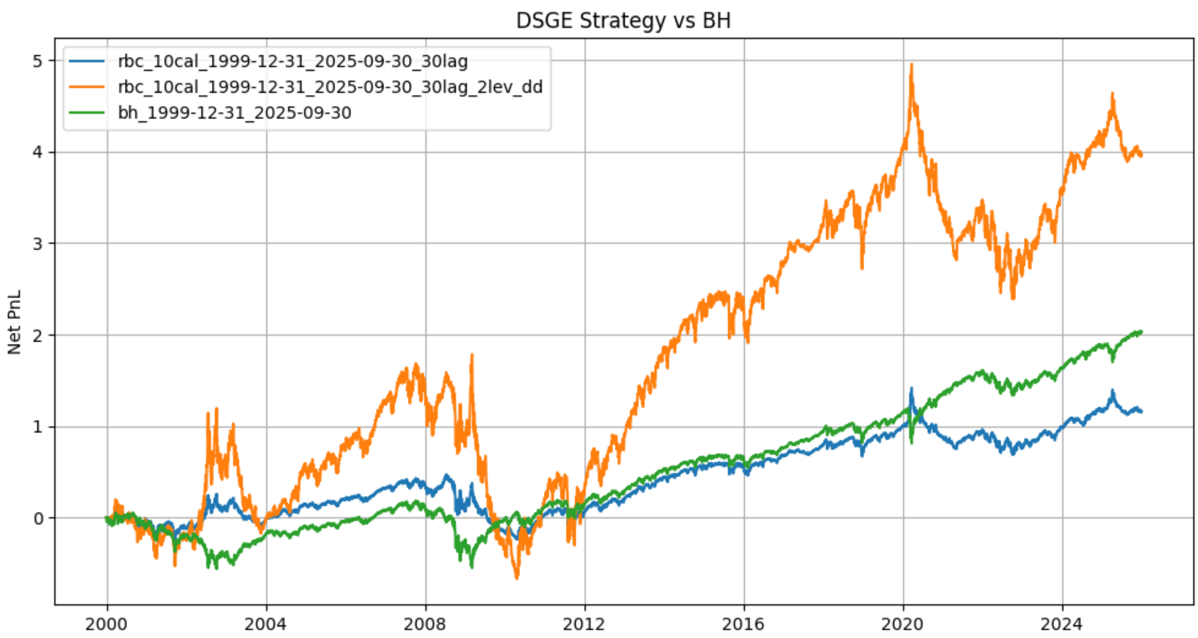

Figure 12: DSGE strategy benchmark with buy-and-hold for S&P 500

Source: own preparation

6. LIMITATIONS AND IMPROVEMENTS

The DSGE trading framework is designed in a way to facilitate the interpretation of DSGE model forecasts and realised macroeconomic data in the context of signal generation in a systematic trading strategy. Whilst confirmed it adds value upon trading, the DSGE trading strategy research questions leave room for future improvements that are explored in this section.

Results show that a basic RBC model and AR1 process estimation are capable of producing the data required for ongoing signal generation. The DSGE strategy in its current form addresses the research questions on containing exploitable information for trading and forecasts being appropriately timed with realised data to generate trading signals. Nevertheless, DSGE modelling enhancements may further improve these aspects. There exist DSGE models that assume the existence of government, export sectors and even financial frictions, each of which would yield different forecast results than the current RBC model (Woodford, 2003; Galí, 2015). Comparison with realised values could then produce different signals. Estimating variables in addition to TFP may also yield improvements, as the DSGE model would benefit from updated model parameters most tailored to the economy in question at a given point in time. Furthermore, DSGE modelling allows the forecasting of variables other than output (such as consumption and investment) whose predictive power could be explored in the context of trading. Such variations may offer alternative trading avenues, producing different signals or ones that pertain to assets other than equity indices. There could also be a different way of applying the DSGE model. The current approach retrains the model on a set data length, produces a next-period forecast, shifts incrementally and continuously repeats the process throughout the horizon. This is driven by the utility of running a DSGE model that is updated with the latest input data and by the utility of near-period forecasts in the IRF. Nevertheless, it may be useful to observe the impact of splitting data into traditional in-sample and out-of-sample periods whereby more IRF quarters are used and the underlying system is not recalibrated past a certain point.

Mean-reversion and momentum are both tested in this framework and they each have their own strengths, particularly when isolating horizons. Analysis in 5.3. Horizon explores the research question of whether forecasts lead or follow realised values in generating profitable trades. 5.3. Horizon also demonstrates when each variant generates the larger PnL and hints at the idea of switching between them based on regime change. Incorporating a trigger for regime change into the framework may constitute a worthy extension of the framework.

A natural extension of the framework following the unchanged SR observation in 5.6. Leverage and Doubling Down would be to introduce modifications to leverage. Research suggests that variance-conditioned dynamic leverage roughly doubles the SR of a static leveraged strategy (Daniel & Moskowitz, 2016).

Finally, assets intuitively associated with the US economy are tested. They are shown to be profitably traded based on signals related to DSGE modelling, addressing the last research question. Broadening the assets tested beyond the S&P 500 and DJIA could result in a wider selection of profitable scenarios. With more macroeconomic variables potentially being able to be forecast by the DSGE model, there could consequently be more asset candidates for trading. For example, a consumption-related equity basket may be traded in response to consumption forecasts.

7. CONCLUSION

The systematic trading strategy introduced in this paper lays the groundwork for a novel type of low-frequency trading based on DSGE modelling. The framework effectively combines DSGE modelling and its forecasts with realised macroeconomic data and systematic trading infrastructure. It processes macroeconomic input data via a functioning RBC model to produce next-period output forecasts and compares these to realised output for generating trading signals. Both momentum and mean-reversion approaches are considered and, alongside several parameter variations, a series of backtests are produced. The entirety of this pipeline is programmatic and reproducible. The creation and execution of the trading strategy produces results that confirm the hypothesis that there exists value in generating DSGE-based forecasts and relating them to realised data for the purpose of trading related assets.

The most successful strategy in terms of performance across the full horizon of data excluding leverage for the S&P 500 asset is the 10 year-calibrated 30 day-signal lagged mean-reversion strategy. It generates a net PnL and net SR slightly lower than the buy-and-hold benchmark, yet is still at healthy levels, is of a similar risk profile having traded infrequently and is designed for long-term holding (due to quarterly position reassessment and positions linked to US macroeconomic data). Results show that a 10-year training period for the TFP series and resulting DSGE model is suitable, as this decreases the chances that several volatile quarters skew regression results and make HP filtering less effective. They also show that a longer 30 business day lag from the reference date of comparison between forecast and realised values to signal implementation is conducive towards increasing PnL, thus reflecting a potential delay between factors driving realised output divergence and equity valuations. The mean-reversion approach proves to be more profitable than the momentum approach overall, indicating realised output growth surprises generally lead forecasts in driving the direction of subsequent corresponding equity moves. This does not hold for all subperiods and could be taken advantage of with additional regime-switching identification techniques.

Leverage in its pure form and doubling down upon position consistency are both effective in increasing PnL. Acknowledging timing constraints and challenges during volatile markets, strategies employing leverage outperform the buy-and-hold benchmark strategy.

The framework produces a healthy low-frequency systematic trading strategy which stands the test of time and is a good alternative to the buy-and-hold strategy, particularly when levered. It confirms the hypothesis and addresses the research questions posed: DSGE models contain exploitable information for trading; DSGE forecasts can be appropriately timed with realised data to generate trading signals; forecasts lead or follow realised values in generating profitable trades; there exist assets that can be profitably traded based on signals related to DSGE modelling. The framework also has the potential to be expanded further in terms of DSGE modelling assumptions, signal generation and assets traded.

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APPENDICES

Appendix A: Framework Setup

Please make sure you have all usual dependencies installed on your system. The most important ones include python, pip and ipykernel. If you're using *Visual Studio Code* for development, you should be prompted to install them automatically. Next, follow this process:

1. Create the virtual environment: python -m venv venv
2. Enter venv: source venv/bin/activate on Linux or venv\Scripts\activate.bat on Windows
3. Upgrade your pip: pip install --upgrade pip
4. Install dependencies from the requirements file: pip install -r requirements.txt
5. Navigate to the Model directory and open the Jupyter Notebook IDE of your choice. Select the Python version from inside the venv you just created when prompted by ipykernel package. Suggested command: python -m notebook

Appendix B: Extended Literature Review

The literature reviewed in this paper spans structural macroeconomic modelling, the relationship between macroeconomic information and asset prices and return predictability over longer horizons. Three articles are particularly relevant to the framework because they address different parts of the mechanism: Christiano, Eichenbaum and Evans (2005) provide a detailed example of how a structural model can generate dynamic responses to an economic shock; Chen, Roll and Ross (1986) provide evidence that macroeconomic innovations are associated with the cross-section of stock returns and Bhojraj and Swaminathan (2006) provide evidence that momentum and subsequent reversal can appear in international equity indices. Taken together, these papers help motivate the combination of DSGE forecasts, macroeconomic surprises and long-horizon equity signals used in this paper.

<u>Christiano, Eichenbaum & Evans (2005)</u>

Christiano, Eichenbaum and Evans (2005) is particularly relevant to the construction of the forecast used in this framework. The paper estimates a structural model of the US economy and studies the dynamic response of macroeconomic variables to a monetary policy shock. The central result is that a model with moderate nominal rigidities can reproduce the inertia in inflation and the persistent, hump-shaped response in output observed after a monetary policy shock. The paper finds that staggered wage contracts are especially important, whilst variable capital utilisation and other real-side mechanisms also contribute to the dynamic response.

The contribution is important for the present framework because it demonstrates why an economic shock should not be treated as a one-period event when translating macroeconomic information into forecasts. The effect of a shock can propagate through the economy over several periods as agents adjust consumption, investment, employment and other decisions. This is conceptually aligned with the use of an impulse response function in the present study. Rather than using the latest productivity observation directly as a trading signal, the framework passes the estimated shock through the RBC model and records the response at a future horizon. The second-horizon output response is then treated as the next-period forecast. The approach is intentionally simpler than the structural model in Christiano et al. (2005), but the underlying principle is similar: economic shocks contain information about a path of future adjustment rather than only the immediate observation.

There are also important differences. Christiano et al. (2005) use an estimated New Keynesian structure with nominal rigidities and monetary policy, whereas the present thesis uses a basic RBC model and a productivity shock. The purpose is therefore not to reproduce their empirical specification, but to adopt the more general forecasting logic of transmitting a structural shock through an equilibrium system. This distinction is important because the relatively simple RBC model limits the set of macroeconomic mechanisms captured by the forecast. It is nevertheless attractive for the current application because the model is computationally light, interpretable and consistent with the quarterly frequency of the input data.

<u>Chen, Roll & Ross (1986)</u>

Chen, Roll and Ross (1986) provide a different but complementary link between macroeconomics and equity returns. Their study tests whether innovations in macroeconomic variables represent systematic risks that are rewarded in the stock market. The variables examined include industrial production growth, changes in expected and unexpected inflation, changes in the risk premium and changes in the term structure of interest rates. The study finds that several of these macroeconomic sources of risk are significantly priced, whilst the market portfolio and aggregate consumption are not separately priced in the same specification. Oil price risk is also not found to provide a separate priced component in the main analysis.

The most important implication for the present framework is that macroeconomic information can have financial relevance even when it is not expressed directly as an asset-price variable. Chen et al. (1986) work with innovations in economic variables and relate their exposures to stock returns. The current strategy takes a related intuition but applies it in a different direction: instead of estimating a static factor exposure between realised macroeconomic innovations and equity returns, it generates a structural forecast of output and compares that forecast with the subsequent realised value. The resulting forecast-realisation gap is therefore an economically interpretable surprise measure that can be converted into a position.

This distinction also clarifies why the present strategy should not be viewed as a conventional factor model. In Chen et al. (1986), the objective is to identify systematic economic sources of risk and test whether those exposures are priced. In the present paper, the objective is to use a model-implied expectation and the deviation from that expectation to determine the direction of a low-frequency trade. The two approaches nevertheless share the idea that macroeconomic information can be connected to asset returns without relying solely on historical price patterns. This supports the choice of output as the macroeconomic variable of interest and of broad US equity indices as the related tradable assets.

Bhojraj & Swaminathan (2006)

Bhojraj and Swaminathan (2006) are directly relevant to the momentum and mean-reversion design of the strategy. Their study examines international equity market indices and finds momentum during approximately the first year following portfolio formation, followed by reversals during the subsequent two years. The authors further show that currency momentum contains information for future stock index returns and can weaken momentum while strengthening later reversals in US dollar-denominated returns. The results are consistent with an interpretation in which markets initially underreact and later correct.

This evidence provides a useful benchmark for interpreting the signal structure used in this thesis. The strategy does not use price momentum in the conventional sense. Instead, it creates momentum and mean-reversion positions from the sign of the difference between DSGE-implied output and realised output. Nevertheless, the economic timing problem is similar. An informational discrepancy can persist for some period, generating continuation, before a later adjustment generates reversal. The results in this thesis show a related pattern: mean-reversion is generally stronger over the full sample, while momentum becomes more useful around the Global Financial Crisis and Covid periods. The framework can therefore be interpreted as testing whether a macroeconomic expectation gap is more likely to lead asset prices or to be followed by a correction, conditional on the economic regime.

The international nature of Bhojraj and Swaminathan (2006) is also useful when considering the asset choice in this thesis. Their results suggest that macroeconomic conditions can be reflected in broad equity indices rather than only in individual securities. The present framework narrows this idea to the S&P 500 and DJIA, which are both broad US equity measures and therefore have a natural economic association with US output. The evidence on reversals also helps explain why a simple comparison of forecast and realised output can generate a persistent signal without requiring daily trading. The signal is slow-moving because the underlying information is quarterly, while the financial response may continue over several subsequent months.

Implications for the present framework

The three articles therefore support different links in the proposed chain from economic model to trade. Christiano et al. (2005) support the use of dynamic shock transmission and impulse responses as a forecasting mechanism. Chen et al. (1986) support the economic relevance of macroeconomic innovations for stock returns. Bhojraj and Swaminathan (2006) support the possibility that macro-linked equity returns can exhibit continuation followed by reversal over extended horizons. The present framework combines these ideas in a novel sequence: a TFP innovation is estimated, passed through an RBC model, translated into an output forecast, compared against realised GDP and then mapped to a long or short position in a related equity index. This combination also highlights the main empirical question of the thesis: whether the forecast-realisation gap contains information that is sufficiently persistent and correctly timed to generate profitable trading signals after accounting for implementation lags and trading costs.

Appendix C: Extended Methods

The methodology in 4. METHODOLOGY can be expressed more formally to clarify the transformations and the timing of the trading signal. The following notation is consistent with the implementation described in the paper and expands the equations without changing the underlying framework.

RBC Model Equilibrium

The setup of the model is available in the dolo Python library from EconForge (Winant, 2019; Winant, 2026).

The full system of equilibrium conditions of the RBC model is listed below.

Production:

$$y_t = e^{z_t} k_t^{\alpha} n_t^{1-\alpha} \tag{9}$$

Resource constraint:

$$c_t + i_t = y_t \tag{10}$$

Capital rental rate:

$$r_t^k = \frac{\alpha y_t}{k_t} \tag{11}$$

Real wage:

$$w_t = \frac{(1-\alpha) y_t}{n_t} \tag{12}$$

Labour supply equilibrium condition:

$$w_t = \chi n_t^{\eta} c_t^{\sigma} \tag{13}$$

Euler equilibrium condition:

$$1 = \beta \left(\frac{c_t}{c_{t+1}}\right)^{\sigma} (1 - \delta + r_{t+1}^k) \tag{14}$$

Productivity law of motion:

$$z_t = \rho z_{t-1} + e_{z,t} \tag{15}$$

Capital accumulation:

$$k_t = (1-\delta) k_{t-1} + i_{t-1} \tag{16}$$

The list of calibrated parameters in the RBC model is shown below.

Discount factor:

$$\beta = 0.99 \tag{17}$$

Capital depreciation rate:

$$\delta = 0.025 \tag{18}$$

Capital share in production:

$$\alpha = 0.33 \tag{19}$$

Curvature of utility with respect to consumption:

$$\sigma = 5 \tag{20}$$

Curvature of the disutility of labour:

$$\eta = 1 \tag{21}$$

Labour disutility scale parameter:

$$\chi = \frac{w}{c^{\sigma} n^{\eta}} \tag{22}$$

TFP process and filtering

Let x_t denote quarterly OPHNFB productivity and t represent quarterly time. The model input is first transformed into logarithms and decomposed into trend and cyclical components using the Hodrick-Prescott filter:

$$\min_{\tau_t} \sum_t (\ln x_t - \tau_t)^2 + \lambda \sum_t [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2, \lambda = 1600 \tag{23}$$

The cyclical component is then approximated by an AR(1) process as in (15).

OLS estimation gives $\hat{\rho}$ and the most recent residual $\hat{e}_{z,t}$. This estimated persistence and latest shock are passed to the RBC model and used to generate the impulse response forecast.

Forecast-realisation gap and signal

Let $\hat{y}_{t|t-1}$ denote the model-implied percentage deviation of output at the selected IRF horizon and y_t the equivalent HP-filtered realised GDP deviation. l is the number of business days by which the signal is lagged. The signal input is the forecast-realisation gap:

$$d_{t+l} = \hat{y}_{t|t-1} - y_t \tag{24}$$

The raw momentum position can be written as $MOM_{t+l} = sign(d_{t+l})$, whilst the mean-reversion position is the inverse, $MR_{t+l} = -sign(d_{t+l})$ with zero applied when $d_{t+l} = 0$. A leverage scalar L_{t+l} is then applied and a doubling-down scalar $D_{t+l} = 2$ is applied when two consecutive quarterly signals have the same direction; otherwise $D_{t+l} = 1$. The implemented position is therefore:

$$p_{t+l} = L_{t+l} D_{t+l} s_{t+l}, s_{t+l} \in \{-1, 0, 1\} \quad (25)$$

<u>Daily PnL and costs</u>

For daily asset prices P_t where t is now daily time, the percentage return is $r_t = \frac{P_t}{P_{t-1}} - 1$. Using the previous day position, gross daily PnL is:

$$g_t = p_{t-1} r_t \quad (26)$$

Net daily PnL subtracts the transaction cost associated with a position change. With κ representing the assumed one-way full-turnover cost, the implementation can be expressed as a proportional turnover term:

$$n_t = g_t - \frac{\kappa}{2} |p_t - p_{t-1}| \quad (27)$$

Cumulative net PnL is the arithmetic sum of n_t over the trading horizon. The annualised SR is obtained from the daily mean and standard deviation of returns across the trading horizon using the 252-trading-day convention:

$$Net\ SR_{annual} = \frac{mean(n_t)}{sd(n_t)} \times \sqrt{252} \quad (28)$$

Appendix D: Extended Results: Strategy Results across Horizons

<u>S&P500 12/1999-09/2025</u>

Model	**Data Length**	**Horizon**	**Signal Lag**	**Leverage**	**Double Down**	**Strategy Type**	**Net PnL**	**Gross PnL**	**Net SR**	**Gross SR**
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.97	3.98	0.24	0.24
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.69	3.7	0.22	0.22
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MR	3.2	3.21	0.19	0.19
RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MR	2.89	2.89	0.17	0.17
RBC	5y	1999-12-31_2025-09-30	10b	2	Yes	MR	2.63	2.64	0.16	0.16
RBC	10y	1999-12-31_2025-09-30	10b	2	Yes	MR	2.47	2.47	0.15	0.15
RBC	10y	1999-12-31_2025-09-30	30b	2	No	MR	2.32	2.33	0.23	0.23
BH	N/A	1999-12-31_2025-09-30	N/A	1	No	N/A	2.03	2.03	0.4	0.4
RBC	10y	1999-12-31_2025-09-30	30b	1	Yes	MR	1.98	1.99	0.24	0.24

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	1999-12-31_2025-09-30	30b	2	No	MR	1.96	1.97	0.2	0.2
RBC	5y	1999-12-31_2025-09-30	30b	1	Yes	MR	1.84	1.85	0.22	0.22
RBC	10y	1999-12-31_2025-09-30	20b	2	No	MR	1.84	1.84	0.18	0.18
RBC	10y	1999-12-31_2025-09-30	20b	1	Yes	MR	1.6	1.6	0.19	0.19
RBC	5y	1999-12-31_2025-09-30	20b	1	Yes	MR	1.44	1.45	0.17	0.17
RBC	5y	1999-12-31_2025-09-30	10b	1	Yes	MR	1.32	1.32	0.16	0.16
RBC	5y	1999-12-31_2025-09-30	20b	2	No	MR	1.27	1.28	0.13	0.13
RBC	10y	1999-12-31_2025-09-30	10b	1	Yes	MR	1.23	1.24	0.15	0.15
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.16	1.16	0.23	0.23
RBC	10y	1999-12-31_2025-09-30	10b	2	No	MR	1.01	1.02	0.1	0.1
RBC	5y	1999-12-31_2025-09-30	30b	1	No	MR	0.98	0.98	0.2	0.2
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MR	0.92	0.92	0.18	0.18
RBC	5y	1999-12-31_2025-09-30	10b	2	No	MR	0.74	0.74	0.07	0.07
RBC	5y	1999-12-31_2025-09-30	20b	1	No	MR	0.64	0.64	0.13	0.13
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MR	0.51	0.51	0.1	0.1
RBC	5y	1999-12-31_2025-09-30	10b	1	No	MR	0.37	0.37	0.07	0.07
RBC	5y	1999-12-31_2025-09-30	10b	1	No	MOM	-0.37	-0.37	-0.07	-0.07
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MOM	-0.51	-0.51	-0.1	-0.1
RBC	5y	1999-12-31_2025-09-30	20b	1	No	MOM	-0.64	-0.64	-0.13	-0.13
RBC	5y	1999-12-31_2025-09-30	10b	2	No	MOM	-0.75	-0.74	-0.07	-0.07
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MOM	-0.92	-0.92	-0.18	-0.18
RBC	5y	1999-12-31_2025-09-30	30b	1	No	MOM	-0.99	-0.98	-0.2	-0.2
RBC	10y	1999-12-31_2025-09-30	10b	2	No	MOM	-1.02	-1.02	-0.1	-0.1
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MOM	-1.17	-1.16	-0.23	-0.23
RBC	10y	1999-12-31_2025-09-30	10b	1	Yes	MOM	-1.24	-1.24	-0.15	-0.15
RBC	5y	1999-12-31_2025-09-30	20b	2	No	MOM	-1.29	-1.28	-0.13	-0.13
RBC	5y	1999-12-31_2025-09-30	10b	1	Yes	MOM	-1.32	-1.32	-0.16	-0.16
RBC	5y	1999-12-31_2025-09-30	20b	1	Yes	MOM	-1.45	-1.45	-0.17	-0.17
RBC	10y	1999-12-31_2025-09-30	20b	1	Yes	MOM	-1.61	-1.6	-0.19	-0.19
RBC	10y	1999-12-31_2025-09-30	20b	2	No	MOM	-1.85	-1.84	-0.18	-0.18
RBC	5y	1999-12-31_2025-09-30	30b	1	Yes	MOM	-1.85	-1.85	-0.22	-0.22
RBC	5y	1999-12-31_2025-09-30	30b	2	No	MOM	-1.97	-1.97	-0.2	-0.2
RBC	10y	1999-12-31_2025-09-30	30b	1	Yes	MOM	-1.99	-1.99	-0.24	-0.24
RBC	10y	1999-12-31_2025-09-30	30b	2	No	MOM	-2.33	-2.33	-0.23	-0.23
RBC	10y	1999-12-31_2025-09-30	10b	2	Yes	MOM	-2.48	-2.47	-0.15	-0.15
RBC	5y	1999-12-31_2025-09-30	10b	2	Yes	MOM	-2.64	-2.64	-0.16	-0.16
RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-2.9	-2.89	-0.17	-0.17

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-3.21	-3.21	-0.19	-0.19
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.7	-3.7	-0.22	-0.22
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.98	-3.98	-0.24	-0.24

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.48	3.48	0.22	0.22
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MR	3.32	3.32	0.21	0.21
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MR	3	3.01	0.19	0.19
RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MR	2.81	2.82	0.18	0.18
RBC	5y	1999-12-31_2025-09-30	10b	2	Yes	MR	2.41	2.42	0.15	0.15
RBC	10y	1999-12-31_2025-09-30	10b	2	Yes	MR	2.18	2.19	0.14	0.14
RBC	10y	1999-12-31_2025-09-30	30b	2	No	MR	2.18	2.18	0.23	0.23
RBC	5y	1999-12-31_2025-09-30	30b	2	No	MR	1.88	1.88	0.2	0.2
BH	N/A	1999-12-31_2025-09-30	N/A	1	No	N/A	1.87	1.87	0.39	0.39
RBC	10y	1999-12-31_2025-09-30	20b	2	No	MR	1.87	1.87	0.2	0.2
RBC	10y	1999-12-31_2025-09-30	30b	1	Yes	MR	1.74	1.74	0.22	0.22
RBC	5y	1999-12-31_2025-09-30	30b	1	Yes	MR	1.66	1.66	0.21	0.21
RBC	10y	1999-12-31_2025-09-30	20b	1	Yes	MR	1.5	1.51	0.19	0.19
RBC	5y	1999-12-31_2025-09-30	20b	1	Yes	MR	1.41	1.41	0.18	0.18
RBC	5y	1999-12-31_2025-09-30	20b	2	No	MR	1.34	1.34	0.14	0.14
RBC	5y	1999-12-31_2025-09-30	10b	1	Yes	MR	1.21	1.21	0.15	0.15
RBC	10y	1999-12-31_2025-09-30	10b	1	Yes	MR	1.09	1.09	0.14	0.14
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.09	1.09	0.23	0.23
RBC	10y	1999-12-31_2025-09-30	10b	2	No	MR	1	1	0.1	0.11
RBC	5y	1999-12-31_2025-09-30	30b	1	No	MR	0.94	0.94	0.2	0.2
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MR	0.93	0.94	0.2	0.2
RBC	5y	1999-12-31_2025-09-30	10b	2	No	MR	0.76	0.77	0.08	0.08
RBC	5y	1999-12-31_2025-09-30	20b	1	No	MR	0.67	0.67	0.14	0.14
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MR	0.5	0.5	0.1	0.11
RBC	5y	1999-12-31_2025-09-30	10b	1	No	MR	0.38	0.38	0.08	0.08
RBC	5y	1999-12-31_2025-09-30	10b	1	No	MOM	-0.38	-0.38	-0.08	-0.08
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MOM	-0.5	-0.5	-0.11	-0.11
RBC	5y	1999-12-31_2025-09-30	20b	1	No	MOM	-0.67	-0.67	-0.14	-0.14
RBC	5y	1999-12-31_2025-09-30	10b	2	No	MOM	-0.77	-0.77	-0.08	-0.08

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MOM	-0.94	-0.94	-0.2	-0.2
RBC	5y	1999-12-31_2025-09-30	30b	1	No	MOM	-0.94	-0.94	-0.2	-0.2
RBC	10y	1999-12-31_2025-09-30	10b	2	No	MOM	-1.01	-1	-0.11	-0.11
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MOM	-1.09	-1.09	-0.23	-0.23
RBC	10y	1999-12-31_2025-09-30	10b	1	Yes	MOM	-1.1	-1.09	-0.14	-0.14
RBC	5y	1999-12-31_2025-09-30	10b	1	Yes	MOM	-1.21	-1.21	-0.15	-0.15
RBC	5y	1999-12-31_2025-09-30	20b	2	No	MOM	-1.35	-1.34	-0.14	-0.14
RBC	5y	1999-12-31_2025-09-30	20b	1	Yes	MOM	-1.41	-1.41	-0.18	-0.18
RBC	10y	1999-12-31_2025-09-30	20b	1	Yes	MOM	-1.51	-1.51	-0.19	-0.19
RBC	5y	1999-12-31_2025-09-30	30b	1	Yes	MOM	-1.67	-1.66	-0.21	-0.21
RBC	10y	1999-12-31_2025-09-30	30b	1	Yes	MOM	-1.74	-1.74	-0.22	-0.22
RBC	10y	1999-12-31_2025-09-30	20b	2	No	MOM	-1.88	-1.87	-0.2	-0.2
RBC	5y	1999-12-31_2025-09-30	30b	2	No	MOM	-1.89	-1.88	-0.2	-0.2
RBC	10y	1999-12-31_2025-09-30	30b	2	No	MOM	-2.19	-2.18	-0.23	-0.23
RBC	10y	1999-12-31_2025-09-30	10b	2	Yes	MOM	-2.19	-2.19	-0.14	-0.14
RBC	5y	1999-12-31_2025-09-30	10b	2	Yes	MOM	-2.43	-2.42	-0.15	-0.15
RBC	5y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-2.83	-2.82	-0.18	-0.18
RBC	10y	1999-12-31_2025-09-30	20b	2	Yes	MOM	-3.02	-3.01	-0.19	-0.19
RBC	5y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.33	-3.32	-0.21	-0.21
RBC	10y	1999-12-31_2025-09-30	30b	2	Yes	MOM	-3.49	-3.48	-0.22	-0.22

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	1999-12-31_2007-12-31	30b	2	Yes	MR	2.24	2.24	0.47	0.47
RBC	5y	1999-12-31_2007-12-31	20b	2	Yes	MR	2.02	2.02	0.42	0.42
RBC	5y	1999-12-31_2007-12-31	10b	2	Yes	MR	1.71	1.71	0.36	0.36
RBC	5y	1999-12-31_2007-12-31	30b	2	No	MR	1.13	1.13	0.39	0.39
RBC	5y	1999-12-31_2007-12-31	30b	1	Yes	MR	1.12	1.12	0.47	0.47
RBC	10y	1999-12-31_2007-12-31	30b	2	Yes	MR	1.06	1.06	0.22	0.22
RBC	5y	1999-12-31_2007-12-31	20b	1	Yes	MR	1.01	1.01	0.42	0.42
RBC	5y	1999-12-31_2007-12-31	20b	2	No	MR	0.96	0.97	0.33	0.33
RBC	5y	1999-12-31_2007-12-31	10b	1	Yes	MR	0.85	0.85	0.36	0.36
RBC	10y	1999-12-31_2007-12-31	20b	2	Yes	MR	0.78	0.78	0.16	0.16
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MR	0.56	0.57	0.39	0.39
RBC	10y	1999-12-31_2007-12-31	30b	2	No	MR	0.54	0.54	0.19	0.19

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2007-12-31	30b	1	Yes	MR	0.53	0.53	0.22	0.22
RBC	5y	1999-12-31_2007-12-31	10b	2	No	MR	0.52	0.52	0.18	0.18
RBC	5y	1999-12-31_2007-12-31	20b	1	No	MR	0.48	0.48	0.33	0.33
RBC	10y	1999-12-31_2007-12-31	20b	1	Yes	MR	0.39	0.39	0.16	0.16
RBC	10y	1999-12-31_2007-12-31	20b	2	No	MR	0.32	0.32	0.11	0.11
RBC	10y	1999-12-31_2007-12-31	10b	2	No	MOM	0.32	0.32	0.11	0.11
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.27	0.27	0.19	0.19
RBC	5y	1999-12-31_2007-12-31	10b	1	No	MR	0.26	0.26	0.18	0.18
RBC	10y	1999-12-31_2007-12-31	10b	2	Yes	MR	0.19	0.19	0.04	0.04
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MR	0.16	0.16	0.11	0.11
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MOM	0.16	0.16	0.11	0.11
RBC	10y	1999-12-31_2007-12-31	10b	1	Yes	MR	0.1	0.1	0.04	0.04
BH	N/A	1999-12-31_2007-12-31	N/A	1	No	N/A	0.03	0.03	0.02	0.02
RBC	10y	1999-12-31_2007-12-31	10b	1	Yes	MOM	-0.1	-0.1	-0.04	-0.04
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MR	-0.16	-0.16	-0.11	-0.11
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MOM	-0.16	-0.16	-0.11	-0.11
RBC	10y	1999-12-31_2007-12-31	10b	2	Yes	MOM	-0.19	-0.19	-0.04	-0.04
RBC	5y	1999-12-31_2007-12-31	10b	1	No	MOM	-0.26	-0.26	-0.18	-0.18
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MOM	-0.27	-0.27	-0.19	-0.19
RBC	10y	1999-12-31_2007-12-31	10b	2	No	MR	-0.32	-0.32	-0.11	-0.11
RBC	10y	1999-12-31_2007-12-31	20b	2	No	MOM	-0.32	-0.32	-0.11	-0.11
RBC	10y	1999-12-31_2007-12-31	20b	1	Yes	MOM	-0.39	-0.39	-0.16	-0.16
RBC	5y	1999-12-31_2007-12-31	20b	1	No	MOM	-0.48	-0.48	-0.33	-0.33
RBC	5y	1999-12-31_2007-12-31	10b	2	No	MOM	-0.53	-0.52	-0.18	-0.18
RBC	10y	1999-12-31_2007-12-31	30b	1	Yes	MOM	-0.53	-0.53	-0.22	-0.22
RBC	10y	1999-12-31_2007-12-31	30b	2	No	MOM	-0.54	-0.54	-0.19	-0.19
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MOM	-0.57	-0.57	-0.39	-0.39
RBC	10y	1999-12-31_2007-12-31	20b	2	Yes	MOM	-0.79	-0.78	-0.16	-0.16
RBC	5y	1999-12-31_2007-12-31	10b	1	Yes	MOM	-0.86	-0.85	-0.36	-0.36
RBC	5y	1999-12-31_2007-12-31	20b	2	No	MOM	-0.97	-0.97	-0.33	-0.33
RBC	5y	1999-12-31_2007-12-31	20b	1	Yes	MOM	-1.01	-1.01	-0.42	-0.42
RBC	10y	1999-12-31_2007-12-31	30b	2	Yes	MOM	-1.06	-1.06	-0.22	-0.22
RBC	5y	1999-12-31_2007-12-31	30b	1	Yes	MOM	-1.12	-1.12	-0.47	-0.47
RBC	5y	1999-12-31_2007-12-31	30b	2	No	MOM	-1.13	-1.13	-0.39	-0.39
RBC	5y	1999-12-31_2007-12-31	10b	2	Yes	MOM	-1.71	-1.71	-0.36	-0.36
RBC	5y	1999-12-31_2007-12-31	20b	2	Yes	MOM	-2.03	-2.02	-0.42	-0.42
RBC	5y	1999-12-31_2007-12-31	30b	2	Yes	MOM	-2.25	-2.24	-0.47	-0.47

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	1999-12-31_2007-12-31	30b	2	Yes	MR	1.94	1.95	0.42	0.42
RBC	5y	1999-12-31_2007-12-31	20b	2	Yes	MR	1.85	1.86	0.4	0.4
RBC	5y	1999-12-31_2007-12-31	10b	2	Yes	MR	1.37	1.37	0.3	0.3
RBC	5y	1999-12-31_2007-12-31	30b	1	Yes	MR	0.97	0.97	0.42	0.42
RBC	5y	1999-12-31_2007-12-31	30b	2	No	MR	0.94	0.94	0.33	0.33
RBC	5y	1999-12-31_2007-12-31	20b	1	Yes	MR	0.93	0.93	0.4	0.4
RBC	5y	1999-12-31_2007-12-31	20b	2	No	MR	0.92	0.92	0.33	0.33
RBC	10y	1999-12-31_2007-12-31	30b	2	Yes	MR	0.81	0.81	0.17	0.17
RBC	5y	1999-12-31_2007-12-31	10b	1	Yes	MR	0.68	0.68	0.3	0.3
RBC	10y	1999-12-31_2007-12-31	20b	2	Yes	MR	0.64	0.64	0.13	0.13
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MR	0.47	0.47	0.33	0.33
RBC	5y	1999-12-31_2007-12-31	20b	1	No	MR	0.46	0.46	0.33	0.33
RBC	10y	1999-12-31_2007-12-31	30b	1	Yes	MR	0.4	0.4	0.17	0.17
RBC	10y	1999-12-31_2007-12-31	10b	2	No	MOM	0.37	0.37	0.13	0.13
RBC	10y	1999-12-31_2007-12-31	30b	2	No	MR	0.36	0.36	0.13	0.13
RBC	5y	1999-12-31_2007-12-31	10b	2	No	MR	0.34	0.34	0.12	0.12
RBC	10y	1999-12-31_2007-12-31	20b	1	Yes	MR	0.32	0.32	0.13	0.13
RBC	10y	1999-12-31_2007-12-31	20b	2	No	MR	0.29	0.29	0.1	0.1
BH	N/A	1999-12-31_2007-12-31	N/A	1	No	N/A	0.19	0.19	0.13	0.13
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MOM	0.19	0.19	0.13	0.13
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.18	0.18	0.13	0.13
RBC	5y	1999-12-31_2007-12-31	10b	1	No	MR	0.17	0.17	0.12	0.12
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MR	0.14	0.14	0.1	0.1
RBC	10y	1999-12-31_2007-12-31	10b	2	Yes	MOM	0	0.01	0	0
RBC	10y	1999-12-31_2007-12-31	10b	1	Yes	MOM	0	0	0	0
RBC	10y	1999-12-31_2007-12-31	10b	1	Yes	MR	0	0	0	0
RBC	10y	1999-12-31_2007-12-31	10b	2	Yes	MR	-0.01	-0.01	0	0
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MOM	-0.15	-0.14	-0.1	-0.1
RBC	5y	1999-12-31_2007-12-31	10b	1	No	MOM	-0.17	-0.17	-0.12	-0.12
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MOM	-0.18	-0.18	-0.13	-0.13
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MR	-0.19	-0.19	-0.13	-0.13
RBC	10y	1999-12-31_2007-12-31	20b	2	No	MOM	-0.29	-0.29	-0.1	-0.1
RBC	10y	1999-12-31_2007-12-31	20b	1	Yes	MOM	-0.32	-0.32	-0.14	-0.13
RBC	5y	1999-12-31_2007-12-31	10b	2	No	MOM	-0.35	-0.34	-0.12	-0.12

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2007-12-31	30b	2	No	MOM	-0.37	-0.36	-0.13	-0.13
RBC	10y	1999-12-31_2007-12-31	10b	2	No	MR	-0.38	-0.37	-0.13	-0.13
RBC	10y	1999-12-31_2007-12-31	30b	1	Yes	MOM	-0.41	-0.4	-0.17	-0.17
RBC	5y	1999-12-31_2007-12-31	20b	1	No	MOM	-0.46	-0.46	-0.33	-0.33
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MOM	-0.47	-0.47	-0.34	-0.33
RBC	10y	1999-12-31_2007-12-31	20b	2	Yes	MOM	-0.64	-0.64	-0.14	-0.13
RBC	5y	1999-12-31_2007-12-31	10b	1	Yes	MOM	-0.69	-0.68	-0.3	-0.3
RBC	10y	1999-12-31_2007-12-31	30b	2	Yes	MOM	-0.81	-0.81	-0.17	-0.17
RBC	5y	1999-12-31_2007-12-31	20b	2	No	MOM	-0.92	-0.92	-0.33	-0.33
RBC	5y	1999-12-31_2007-12-31	20b	1	Yes	MOM	-0.93	-0.93	-0.4	-0.4
RBC	5y	1999-12-31_2007-12-31	30b	2	No	MOM	-0.94	-0.94	-0.34	-0.33
RBC	5y	1999-12-31_2007-12-31	30b	1	Yes	MOM	-0.97	-0.97	-0.42	-0.42
RBC	5y	1999-12-31_2007-12-31	10b	2	Yes	MOM	-1.37	-1.37	-0.3	-0.3
RBC	5y	1999-12-31_2007-12-31	20b	2	Yes	MOM	-1.86	-1.86	-0.4	-0.4
RBC	5y	1999-12-31_2007-12-31	30b	2	Yes	MOM	-1.95	-1.95	-0.42	-0.42

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2007-12-31_2009-12-31	10b	2	Yes	MOM	1.8	1.8	0.88	0.88
RBC	10y	2007-12-31_2009-12-31	30b	2	Yes	MOM	1.75	1.75	0.89	0.89
RBC	5y	2007-12-31_2009-12-31	30b	2	Yes	MOM	1.72	1.72	0.84	0.84
RBC	10y	2007-12-31_2009-12-31	20b	2	Yes	MOM	1.62	1.62	0.81	0.81
RBC	5y	2007-12-31_2009-12-31	20b	2	Yes	MOM	1.57	1.57	0.76	0.76
RBC	5y	2007-12-31_2009-12-31	10b	2	Yes	MOM	1.51	1.51	0.71	0.71
RBC	5y	2007-12-31_2009-12-31	30b	2	No	MOM	1.12	1.12	0.76	0.76
RBC	10y	2007-12-31_2009-12-31	10b	2	No	MOM	1.08	1.08	0.72	0.72
RBC	5y	2007-12-31_2009-12-31	20b	2	No	MOM	1	1	0.68	0.68
RBC	10y	2007-12-31_2009-12-31	30b	2	No	MOM	0.99	0.99	0.67	0.67
RBC	5y	2007-12-31_2009-12-31	10b	2	No	MOM	0.95	0.95	0.64	0.64
RBC	10y	2007-12-31_2009-12-31	10b	1	Yes	MOM	0.9	0.9	0.88	0.88
RBC	10y	2007-12-31_2009-12-31	30b	1	Yes	MOM	0.87	0.87	0.89	0.89
RBC	10y	2007-12-31_2009-12-31	20b	2	No	MOM	0.86	0.86	0.58	0.58
RBC	5y	2007-12-31_2009-12-31	30b	1	Yes	MOM	0.86	0.86	0.84	0.84
RBC	10y	2007-12-31_2009-12-31	20b	1	Yes	MOM	0.81	0.81	0.81	0.81
RBC	5y	2007-12-31_2009-12-31	20b	1	Yes	MOM	0.79	0.79	0.76	0.76

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	2007-12-31_2009-12-31	10b	1	Yes	MOM	0.75	0.75	0.71	0.71
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MOM	0.56	0.56	0.76	0.76
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MOM	0.54	0.54	0.72	0.72
RBC	5y	2007-12-31_2009-12-31	20b	1	No	MOM	0.5	0.5	0.68	0.68
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MOM	0.5	0.5	0.67	0.67
RBC	5y	2007-12-31_2009-12-31	10b	1	No	MOM	0.47	0.47	0.64	0.64
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MOM	0.43	0.43	0.58	0.58
BH	N/A	2007-12-31_2009-12-31	N/A	1	No	N/A	-0.1	-0.1	-0.14	-0.14
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MR	-0.43	-0.43	-0.58	-0.58
RBC	5y	2007-12-31_2009-12-31	10b	1	No	MR	-0.47	-0.47	-0.64	-0.64
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.5	-0.5	-0.67	-0.67
RBC	5y	2007-12-31_2009-12-31	20b	1	No	MR	-0.5	-0.5	-0.68	-0.68
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MR	-0.54	-0.54	-0.72	-0.72
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MR	-0.56	-0.56	-0.76	-0.76
RBC	5y	2007-12-31_2009-12-31	10b	1	Yes	MR	-0.75	-0.75	-0.71	-0.71
RBC	5y	2007-12-31_2009-12-31	20b	1	Yes	MR	-0.79	-0.79	-0.76	-0.76
RBC	10y	2007-12-31_2009-12-31	20b	1	Yes	MR	-0.81	-0.81	-0.81	-0.81
RBC	5y	2007-12-31_2009-12-31	30b	1	Yes	MR	-0.86	-0.86	-0.84	-0.84
RBC	10y	2007-12-31_2009-12-31	20b	2	No	MR	-0.86	-0.86	-0.58	-0.58
RBC	10y	2007-12-31_2009-12-31	30b	1	Yes	MR	-0.87	-0.87	-0.89	-0.89
RBC	10y	2007-12-31_2009-12-31	10b	1	Yes	MR	-0.9	-0.9	-0.88	-0.88
RBC	5y	2007-12-31_2009-12-31	10b	2	No	MR	-0.95	-0.95	-0.64	-0.64
RBC	10y	2007-12-31_2009-12-31	30b	2	No	MR	-0.99	-0.99	-0.67	-0.67
RBC	5y	2007-12-31_2009-12-31	20b	2	No	MR	-1	-1	-0.68	-0.68
RBC	10y	2007-12-31_2009-12-31	10b	2	No	MR	-1.08	-1.08	-0.72	-0.72
RBC	5y	2007-12-31_2009-12-31	30b	2	No	MR	-1.12	-1.12	-0.76	-0.76
RBC	5y	2007-12-31_2009-12-31	10b	2	Yes	MR	-1.51	-1.51	-0.71	-0.71
RBC	5y	2007-12-31_2009-12-31	20b	2	Yes	MR	-1.58	-1.57	-0.76	-0.76
RBC	10y	2007-12-31_2009-12-31	20b	2	Yes	MR	-1.62	-1.62	-0.81	-0.81
RBC	5y	2007-12-31_2009-12-31	30b	2	Yes	MR	-1.72	-1.72	-0.84	-0.84
RBC	10y	2007-12-31_2009-12-31	30b	2	Yes	MR	-1.75	-1.75	-0.89	-0.89
RBC	10y	2007-12-31_2009-12-31	10b	2	Yes	MR	-1.81	-1.8	-0.88	-0.88

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2007-12-31_2009-12-31	10b	2	Yes	MOM	1.45	1.45	0.78	0.78
RBC	10y	2007-12-31_2009-12-31	30b	2	Yes	MOM	1.42	1.42	0.8	0.8
RBC	5y	2007-12-31_2009-12-31	30b	2	Yes	MOM	1.35	1.35	0.72	0.72
RBC	10y	2007-12-31_2009-12-31	20b	2	Yes	MOM	1.24	1.24	0.68	0.68
RBC	5y	2007-12-31_2009-12-31	10b	2	Yes	MOM	1.19	1.19	0.62	0.62
RBC	5y	2007-12-31_2009-12-31	20b	2	Yes	MOM	1.16	1.16	0.62	0.62
RBC	5y	2007-12-31_2009-12-31	30b	2	No	MOM	0.84	0.84	0.63	0.63
RBC	10y	2007-12-31_2009-12-31	10b	2	No	MOM	0.83	0.83	0.61	0.62
RBC	5y	2007-12-31_2009-12-31	10b	2	No	MOM	0.79	0.79	0.58	0.58
RBC	5y	2007-12-31_2009-12-31	20b	2	No	MOM	0.73	0.73	0.54	0.54
RBC	10y	2007-12-31_2009-12-31	10b	1	Yes	MOM	0.73	0.73	0.78	0.78
RBC	10y	2007-12-31_2009-12-31	30b	2	No	MOM	0.72	0.72	0.54	0.54
RBC	10y	2007-12-31_2009-12-31	30b	1	Yes	MOM	0.71	0.71	0.8	0.8
RBC	5y	2007-12-31_2009-12-31	30b	1	Yes	MOM	0.67	0.68	0.72	0.72
RBC	10y	2007-12-31_2009-12-31	20b	1	Yes	MOM	0.62	0.62	0.68	0.68
RBC	5y	2007-12-31_2009-12-31	10b	1	Yes	MOM	0.59	0.59	0.62	0.62
RBC	5y	2007-12-31_2009-12-31	20b	1	Yes	MOM	0.58	0.58	0.62	0.62
RBC	10y	2007-12-31_2009-12-31	20b	2	No	MOM	0.57	0.57	0.42	0.42
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MOM	0.42	0.42	0.63	0.63
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MOM	0.42	0.42	0.61	0.62
RBC	5y	2007-12-31_2009-12-31	10b	1	No	MOM	0.4	0.4	0.58	0.58
RBC	5y	2007-12-31_2009-12-31	20b	1	No	MOM	0.36	0.37	0.54	0.54
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MOM	0.36	0.36	0.54	0.54
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MOM	0.29	0.29	0.42	0.42
BH	N/A	2007-12-31_2009-12-31	N/A	1	No	N/A	-0.1	-0.1	-0.14	-0.14
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MR	-0.29	-0.29	-0.43	-0.42
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.36	-0.36	-0.54	-0.54
RBC	5y	2007-12-31_2009-12-31	20b	1	No	MR	-0.37	-0.37	-0.54	-0.54
RBC	5y	2007-12-31_2009-12-31	10b	1	No	MR	-0.4	-0.4	-0.59	-0.58
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MR	-0.42	-0.42	-0.62	-0.62
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MR	-0.42	-0.42	-0.63	-0.63
RBC	10y	2007-12-31_2009-12-31	20b	2	No	MR	-0.57	-0.57	-0.43	-0.42
RBC	5y	2007-12-31_2009-12-31	20b	1	Yes	MR	-0.58	-0.58	-0.62	-0.62
RBC	5y	2007-12-31_2009-12-31	10b	1	Yes	MR	-0.59	-0.59	-0.62	-0.62
RBC	10y	2007-12-31_2009-12-31	20b	1	Yes	MR	-0.62	-0.62	-0.68	-0.68
RBC	5y	2007-12-31_2009-12-31	30b	1	Yes	MR	-0.68	-0.68	-0.72	-0.72
RBC	10y	2007-12-31_2009-12-31	30b	1	Yes	MR	-0.71	-0.71	-0.8	-0.8

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2007-12-31_2009-12-31	30b	2	No	MR	-0.72	-0.72	-0.54	-0.54
RBC	10y	2007-12-31_2009-12-31	10b	1	Yes	MR	-0.73	-0.73	-0.78	-0.78
RBC	5y	2007-12-31_2009-12-31	20b	2	No	MR	-0.73	-0.73	-0.54	-0.54
RBC	5y	2007-12-31_2009-12-31	10b	2	No	MR	-0.79	-0.79	-0.59	-0.58
RBC	10y	2007-12-31_2009-12-31	10b	2	No	MR	-0.84	-0.83	-0.62	-0.62
RBC	5y	2007-12-31_2009-12-31	30b	2	No	MR	-0.84	-0.84	-0.63	-0.63
RBC	5y	2007-12-31_2009-12-31	20b	2	Yes	MR	-1.16	-1.16	-0.62	-0.62
RBC	5y	2007-12-31_2009-12-31	10b	2	Yes	MR	-1.19	-1.19	-0.62	-0.62
RBC	10y	2007-12-31_2009-12-31	20b	2	Yes	MR	-1.24	-1.24	-0.68	-0.68
RBC	5y	2007-12-31_2009-12-31	30b	2	Yes	MR	-1.35	-1.35	-0.72	-0.72
RBC	10y	2007-12-31_2009-12-31	30b	2	Yes	MR	-1.42	-1.42	-0.8	-0.8
RBC	10y	2007-12-31_2009-12-31	10b	2	Yes	MR	-1.45	-1.45	-0.78	-0.78

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2009-12-31_2019-12-31	10b	2	Yes	MR	5.19	5.19	0.85	0.85
RBC	10y	2009-12-31_2019-12-31	30b	2	Yes	MR	5.04	5.04	0.83	0.83
RBC	20y	2009-12-31_2019-12-31	10b	2	Yes	MR	5.02	5.02	0.83	0.83
RBC	10y	2009-12-31_2019-12-31	20b	2	Yes	MR	4.81	4.81	0.79	0.79
RBC	20y	2009-12-31_2019-12-31	20b	2	Yes	MR	4.6	4.6	0.77	0.77
RBC	20y	2009-12-31_2019-12-31	30b	2	Yes	MR	4.43	4.43	0.74	0.74
RBC	5y	2009-12-31_2019-12-31	30b	2	Yes	MR	4.17	4.17	0.7	0.71
RBC	5y	2009-12-31_2019-12-31	10b	2	Yes	MR	4.14	4.14	0.7	0.7
RBC	5y	2009-12-31_2019-12-31	20b	2	Yes	MR	3.95	3.95	0.67	0.67
RBC	20y	2009-12-31_2019-12-31	10b	2	No	MR	2.86	2.86	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	10b	2	No	MR	2.85	2.85	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	30b	2	No	MR	2.79	2.79	0.81	0.81
RBC	10y	2009-12-31_2019-12-31	10b	1	Yes	MR	2.6	2.6	0.85	0.85
RBC	10y	2009-12-31_2019-12-31	20b	2	No	MR	2.59	2.59	0.75	0.75
RBC	20y	2009-12-31_2019-12-31	20b	2	No	MR	2.56	2.56	0.74	0.74
RBC	10y	2009-12-31_2019-12-31	30b	1	Yes	MR	2.52	2.52	0.83	0.83
RBC	20y	2009-12-31_2019-12-31	10b	1	Yes	MR	2.51	2.51	0.83	0.83
RBC	20y	2009-12-31_2019-12-31	30b	2	No	MR	2.5	2.5	0.72	0.72
RBC	10y	2009-12-31_2019-12-31	20b	1	Yes	MR	2.4	2.4	0.79	0.79
RBC	5y	2009-12-31_2019-12-31	30b	2	No	MR	2.36	2.37	0.68	0.68

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	20y	2009-12-31_2019-12-31	20b	1	Yes	MR	2.3	2.3	0.77	0.77
RBC	20y	2009-12-31_2019-12-31	30b	1	Yes	MR	2.21	2.21	0.74	0.74
RBC	5y	2009-12-31_2019-12-31	10b	2	No	MR	2.12	2.12	0.61	0.61
RBC	5y	2009-12-31_2019-12-31	20b	2	No	MR	2.1	2.1	0.6	0.6
RBC	5y	2009-12-31_2019-12-31	30b	1	Yes	MR	2.08	2.09	0.7	0.71
RBC	5y	2009-12-31_2019-12-31	10b	1	Yes	MR	2.07	2.07	0.7	0.7
RBC	5y	2009-12-31_2019-12-31	20b	1	Yes	MR	1.98	1.98	0.67	0.67
RBC	20y	2009-12-31_2019-12-31	10b	1	No	MR	1.43	1.43	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MR	1.42	1.42	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.4	1.4	0.81	0.81
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MR	1.3	1.3	0.75	0.75
RBC	20y	2009-12-31_2019-12-31	20b	1	No	MR	1.28	1.28	0.74	0.74
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MR	1.25	1.25	0.72	0.72
RBC	5y	2009-12-31_2019-12-31	30b	1	No	MR	1.18	1.18	0.68	0.68
RBC	5y	2009-12-31_2019-12-31	10b	1	No	MR	1.06	1.06	0.61	0.61
RBC	5y	2009-12-31_2019-12-31	20b	1	No	MR	1.05	1.05	0.6	0.6
BH	N/A	2009-12-31_2019-12-31	N/A	1	No	N/A	0.99	0.99	0.57	0.57
RBC	5y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.05	-1.05	-0.6	-0.6
RBC	5y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.06	-1.06	-0.61	-0.61
RBC	5y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.18	-1.18	-0.68	-0.68
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.25	-1.25	-0.72	-0.72
RBC	20y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.28	-1.28	-0.74	-0.74
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.3	-1.3	-0.75	-0.75
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.4	-1.4	-0.81	-0.81
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.42	-1.42	-0.82	-0.82
RBC	20y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.43	-1.43	-0.82	-0.82
RBC	5y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-1.98	-1.98	-0.67	-0.67
RBC	5y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-2.07	-2.07	-0.7	-0.7
RBC	5y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-2.09	-2.09	-0.71	-0.71
RBC	5y	2009-12-31_2019-12-31	20b	2	No	MOM	-2.1	-2.1	-0.6	-0.6
RBC	5y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.12	-2.12	-0.61	-0.61
RBC	20y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-2.22	-2.21	-0.74	-0.74
RBC	20y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-2.3	-2.3	-0.77	-0.77
RBC	5y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.37	-2.37	-0.68	-0.68
RBC	10y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-2.4	-2.4	-0.79	-0.79
RBC	20y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.5	-2.5	-0.72	-0.72
RBC	20y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-2.51	-2.51	-0.84	-0.83

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-2.52	-2.52	-0.83	-0.83
RBC	20y	2009-12-31_2019-12-31	20b	2	No	MOM	-2.56	-2.56	-0.74	-0.74
RBC	10y	2009-12-31_2019-12-31	20b	2	No	MOM	-2.59	-2.59	-0.75	-0.75
RBC	10y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-2.6	-2.6	-0.85	-0.85
RBC	10y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.79	-2.79	-0.81	-0.81
RBC	10y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.85	-2.85	-0.82	-0.82
RBC	20y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.86	-2.86	-0.82	-0.82
RBC	5y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-3.96	-3.95	-0.67	-0.67
RBC	5y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-4.14	-4.14	-0.7	-0.7
RBC	5y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-4.18	-4.17	-0.71	-0.71
RBC	20y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-4.43	-4.43	-0.74	-0.74
RBC	20y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-4.6	-4.6	-0.77	-0.77
RBC	10y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-4.81	-4.81	-0.79	-0.79
RBC	20y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-5.02	-5.02	-0.84	-0.83
RBC	10y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-5.04	-5.04	-0.83	-0.83
RBC	10y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-5.2	-5.19	-0.85	-0.85

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2009-12-31_2019-12-31	10b	2	Yes	MR	4.86	4.86	0.82	0.82
RBC	20y	2009-12-31_2019-12-31	10b	2	Yes	MR	4.77	4.77	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	30b	2	Yes	MR	4.51	4.52	0.77	0.77
RBC	10y	2009-12-31_2019-12-31	20b	2	Yes	MR	4.37	4.37	0.74	0.74
RBC	20y	2009-12-31_2019-12-31	20b	2	Yes	MR	4.25	4.26	0.73	0.73
RBC	20y	2009-12-31_2019-12-31	30b	2	Yes	MR	3.96	3.96	0.69	0.69
RBC	5y	2009-12-31_2019-12-31	10b	2	Yes	MR	3.86	3.87	0.67	0.67
RBC	5y	2009-12-31_2019-12-31	30b	2	Yes	MR	3.78	3.78	0.66	0.66
RBC	5y	2009-12-31_2019-12-31	20b	2	Yes	MR	3.62	3.62	0.63	0.63
RBC	20y	2009-12-31_2019-12-31	10b	2	No	MR	2.78	2.78	0.81	0.81
RBC	10y	2009-12-31_2019-12-31	10b	2	No	MR	2.74	2.74	0.8	0.8
RBC	10y	2009-12-31_2019-12-31	30b	2	No	MR	2.54	2.54	0.74	0.74
RBC	10y	2009-12-31_2019-12-31	10b	1	Yes	MR	2.43	2.43	0.82	0.82
RBC	20y	2009-12-31_2019-12-31	20b	2	No	MR	2.42	2.43	0.71	0.71
RBC	10y	2009-12-31_2019-12-31	20b	2	No	MR	2.41	2.42	0.7	0.71
RBC	20y	2009-12-31_2019-12-31	10b	1	Yes	MR	2.38	2.38	0.82	0.82

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2009-12-31_2019-12-31	30b	1	Yes	MR	2.26	2.26	0.77	0.77
RBC	20y	2009-12-31_2019-12-31	30b	2	No	MR	2.25	2.25	0.66	0.66
RBC	5y	2009-12-31_2019-12-31	30b	2	No	MR	2.2	2.2	0.64	0.64
RBC	10y	2009-12-31_2019-12-31	20b	1	Yes	MR	2.19	2.19	0.74	0.74
RBC	20y	2009-12-31_2019-12-31	20b	1	Yes	MR	2.13	2.13	0.73	0.73
RBC	5y	2009-12-31_2019-12-31	10b	2	No	MR	2.06	2.06	0.6	0.6
RBC	20y	2009-12-31_2019-12-31	30b	1	Yes	MR	1.98	1.98	0.69	0.69
RBC	5y	2009-12-31_2019-12-31	20b	2	No	MR	1.97	1.97	0.57	0.58
RBC	5y	2009-12-31_2019-12-31	10b	1	Yes	MR	1.93	1.93	0.67	0.67
RBC	5y	2009-12-31_2019-12-31	30b	1	Yes	MR	1.89	1.89	0.66	0.66
RBC	5y	2009-12-31_2019-12-31	20b	1	Yes	MR	1.81	1.81	0.63	0.63
RBC	20y	2009-12-31_2019-12-31	10b	1	No	MR	1.39	1.39	0.81	0.81
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MR	1.37	1.37	0.8	0.8
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.27	1.27	0.74	0.74
RBC	20y	2009-12-31_2019-12-31	20b	1	No	MR	1.21	1.21	0.71	0.71
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MR	1.21	1.21	0.7	0.71
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MR	1.12	1.12	0.66	0.66
RBC	5y	2009-12-31_2019-12-31	30b	1	No	MR	1.1	1.1	0.64	0.64
RBC	5y	2009-12-31_2019-12-31	10b	1	No	MR	1.03	1.03	0.6	0.6
RBC	5y	2009-12-31_2019-12-31	20b	1	No	MR	0.98	0.99	0.57	0.58
BH	N/A	2009-12-31_2019-12-31	N/A	1	No	N/A	0.89	0.89	0.52	0.52
RBC	5y	2009-12-31_2019-12-31	20b	1	No	MOM	-0.99	-0.99	-0.58	-0.58
RBC	5y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.03	-1.03	-0.6	-0.6
RBC	5y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.1	-1.1	-0.65	-0.64
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.13	-1.12	-0.66	-0.66
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.21	-1.21	-0.71	-0.71
RBC	20y	2009-12-31_2019-12-31	20b	1	No	MOM	-1.21	-1.21	-0.71	-0.71
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MOM	-1.27	-1.27	-0.74	-0.74
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.37	-1.37	-0.8	-0.8
RBC	20y	2009-12-31_2019-12-31	10b	1	No	MOM	-1.39	-1.39	-0.81	-0.81
RBC	5y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-1.81	-1.81	-0.63	-0.63
RBC	5y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-1.89	-1.89	-0.66	-0.66
RBC	5y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-1.93	-1.93	-0.67	-0.67
RBC	5y	2009-12-31_2019-12-31	20b	2	No	MOM	-1.97	-1.97	-0.58	-0.58
RBC	20y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-1.98	-1.98	-0.69	-0.69
RBC	5y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.06	-2.06	-0.6	-0.6
RBC	20y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-2.13	-2.13	-0.73	-0.73

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2009-12-31_2019-12-31	20b	1	Yes	MOM	-2.19	-2.19	-0.74	-0.74
RBC	5y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.21	-2.2	-0.65	-0.64
RBC	20y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.25	-2.25	-0.66	-0.66
RBC	10y	2009-12-31_2019-12-31	30b	1	Yes	MOM	-2.26	-2.26	-0.77	-0.77
RBC	20y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-2.38	-2.38	-0.82	-0.82
RBC	10y	2009-12-31_2019-12-31	20b	2	No	MOM	-2.42	-2.42	-0.71	-0.71
RBC	20y	2009-12-31_2019-12-31	20b	2	No	MOM	-2.43	-2.43	-0.71	-0.71
RBC	10y	2009-12-31_2019-12-31	10b	1	Yes	MOM	-2.43	-2.43	-0.82	-0.82
RBC	10y	2009-12-31_2019-12-31	30b	2	No	MOM	-2.54	-2.54	-0.74	-0.74
RBC	10y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.74	-2.74	-0.8	-0.8
RBC	20y	2009-12-31_2019-12-31	10b	2	No	MOM	-2.78	-2.78	-0.81	-0.81
RBC	5y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-3.63	-3.62	-0.63	-0.63
RBC	5y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-3.78	-3.78	-0.66	-0.66
RBC	5y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-3.87	-3.87	-0.67	-0.67
RBC	20y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-3.96	-3.96	-0.69	-0.69
RBC	20y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-4.26	-4.26	-0.73	-0.73
RBC	10y	2009-12-31_2019-12-31	20b	2	Yes	MOM	-4.37	-4.37	-0.74	-0.74
RBC	10y	2009-12-31_2019-12-31	30b	2	Yes	MOM	-4.52	-4.52	-0.77	-0.77
RBC	20y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-4.77	-4.77	-0.82	-0.82
RBC	10y	2009-12-31_2019-12-31	10b	2	Yes	MOM	-4.86	-4.86	-0.82	-0.82

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.43	1.44	0.55	0.55
RBC	10y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.43	1.44	0.55	0.55
RBC	20y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.43	1.44	0.55	0.55
RBC	5y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.33	1.33	0.51	0.51
RBC	10y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.33	1.33	0.51	0.51
RBC	20y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.33	1.33	0.51	0.51
RBC	5y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.22	1.22	0.47	0.47
RBC	10y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.22	1.22	0.47	0.47
RBC	20y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.22	1.22	0.47	0.47
RBC	5y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.72	0.72	0.55	0.55
RBC	10y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.72	0.72	0.55	0.55
RBC	20y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.72	0.72	0.55	0.55

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.67	0.67	0.51	0.51
RBC	10y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.67	0.67	0.51	0.51
RBC	20y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.67	0.67	0.51	0.51
RBC	5y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.61	0.61	0.47	0.47
RBC	10y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.61	0.61	0.47	0.47
RBC	20y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.61	0.61	0.47	0.47
RBC	5y	2019-12-31_2022-12-31	20b	2	No	MOM	0.57	0.57	0.35	0.35
RBC	10y	2019-12-31_2022-12-31	20b	2	No	MOM	0.57	0.57	0.35	0.35
RBC	20y	2019-12-31_2022-12-31	20b	2	No	MOM	0.57	0.57	0.35	0.35
RBC	5y	2019-12-31_2022-12-31	10b	2	No	MOM	0.51	0.51	0.32	0.32
RBC	10y	2019-12-31_2022-12-31	10b	2	No	MOM	0.51	0.51	0.32	0.32
RBC	20y	2019-12-31_2022-12-31	10b	2	No	MOM	0.51	0.51	0.32	0.32
RBC	5y	2019-12-31_2022-12-31	30b	2	No	MOM	0.4	0.4	0.25	0.25
RBC	10y	2019-12-31_2022-12-31	30b	2	No	MOM	0.4	0.4	0.25	0.25
RBC	20y	2019-12-31_2022-12-31	30b	2	No	MOM	0.4	0.4	0.25	0.25
BH	N/A	2019-12-31_2022-12-31	N/A	1	No	N/A	0.34	0.34	0.42	0.42
RBC	5y	2019-12-31_2022-12-31	20b	1	No	MOM	0.29	0.29	0.35	0.35
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MOM	0.29	0.29	0.35	0.35
RBC	20y	2019-12-31_2022-12-31	20b	1	No	MOM	0.29	0.29	0.35	0.35
RBC	5y	2019-12-31_2022-12-31	10b	1	No	MOM	0.25	0.26	0.32	0.32
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MOM	0.25	0.26	0.32	0.32
RBC	20y	2019-12-31_2022-12-31	10b	1	No	MOM	0.25	0.26	0.32	0.32
RBC	5y	2019-12-31_2022-12-31	30b	1	No	MOM	0.2	0.2	0.25	0.25
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MOM	0.2	0.2	0.25	0.25
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MOM	0.2	0.2	0.25	0.25
RBC	5y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	5y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	20y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	5y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	20y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	5y	2019-12-31_2022-12-31	30b	2	No	MR	-0.4	-0.4	-0.25	-0.25
RBC	10y	2019-12-31_2022-12-31	30b	2	No	MR	-0.4	-0.4	-0.25	-0.25
RBC	20y	2019-12-31_2022-12-31	30b	2	No	MR	-0.4	-0.4	-0.25	-0.25

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	2019-12-31_2022-12-31	10b	2	No	MR	-0.51	-0.51	-0.32	-0.32
RBC	10y	2019-12-31_2022-12-31	10b	2	No	MR	-0.51	-0.51	-0.32	-0.32
RBC	20y	2019-12-31_2022-12-31	10b	2	No	MR	-0.51	-0.51	-0.32	-0.32
RBC	5y	2019-12-31_2022-12-31	20b	2	No	MR	-0.57	-0.57	-0.35	-0.35
RBC	10y	2019-12-31_2022-12-31	20b	2	No	MR	-0.57	-0.57	-0.35	-0.35
RBC	20y	2019-12-31_2022-12-31	20b	2	No	MR	-0.57	-0.57	-0.35	-0.35
RBC	5y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.61	-0.61	-0.47	-0.47
RBC	10y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.61	-0.61	-0.47	-0.47
RBC	20y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.61	-0.61	-0.47	-0.47
RBC	5y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.67	-0.67	-0.51	-0.51
RBC	10y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.67	-0.67	-0.51	-0.51
RBC	20y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.67	-0.67	-0.51	-0.51
RBC	5y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.72	-0.72	-0.55	-0.55
RBC	10y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.72	-0.72	-0.55	-0.55
RBC	20y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.72	-0.72	-0.55	-0.55
RBC	5y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.22	-1.22	-0.47	-0.47
RBC	10y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.22	-1.22	-0.47	-0.47
RBC	20y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.22	-1.22	-0.47	-0.47
RBC	5y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.33	-1.33	-0.51	-0.51
RBC	10y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.33	-1.33	-0.51	-0.51
RBC	20y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.33	-1.33	-0.51	-0.51
RBC	5y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.44	-1.44	-0.55	-0.55
RBC	10y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.44	-1.44	-0.55	-0.55
RBC	20y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.44	-1.44	-0.55	-0.55

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.2	1.2	0.5	0.5
RBC	10y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.2	1.2	0.5	0.5
RBC	20y	2019-12-31_2022-12-31	20b	2	Yes	MOM	1.2	1.2	0.5	0.5
RBC	5y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.18	1.19	0.48	0.48
RBC	10y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.18	1.19	0.48	0.48
RBC	20y	2019-12-31_2022-12-31	10b	2	Yes	MOM	1.18	1.19	0.48	0.48
RBC	5y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.17	1.18	0.49	0.49
RBC	10y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.17	1.18	0.49	0.49

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	20y	2019-12-31_2022-12-31	30b	2	Yes	MOM	1.17	1.18	0.49	0.49
RBC	5y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.6	0.6	0.5	0.5
RBC	10y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.6	0.6	0.5	0.5
RBC	20y	2019-12-31_2022-12-31	20b	1	Yes	MOM	0.6	0.6	0.5	0.5
RBC	5y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.59	0.59	0.48	0.48
RBC	10y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.59	0.59	0.48	0.48
RBC	20y	2019-12-31_2022-12-31	10b	1	Yes	MOM	0.59	0.59	0.48	0.48
RBC	5y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.59	0.59	0.49	0.49
RBC	10y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.59	0.59	0.49	0.49
RBC	20y	2019-12-31_2022-12-31	30b	1	Yes	MOM	0.59	0.59	0.49	0.49
RBC	5y	2019-12-31_2022-12-31	20b	2	No	MOM	0.44	0.44	0.27	0.27
RBC	10y	2019-12-31_2022-12-31	20b	2	No	MOM	0.44	0.44	0.27	0.27
RBC	20y	2019-12-31_2022-12-31	20b	2	No	MOM	0.44	0.44	0.27	0.27
RBC	5y	2019-12-31_2022-12-31	10b	2	No	MOM	0.41	0.41	0.26	0.26
RBC	10y	2019-12-31_2022-12-31	10b	2	No	MOM	0.41	0.41	0.26	0.26
RBC	20y	2019-12-31_2022-12-31	10b	2	No	MOM	0.41	0.41	0.26	0.26
RBC	5y	2019-12-31_2022-12-31	30b	2	No	MOM	0.37	0.37	0.23	0.23
RBC	10y	2019-12-31_2022-12-31	30b	2	No	MOM	0.37	0.37	0.23	0.23
RBC	20y	2019-12-31_2022-12-31	30b	2	No	MOM	0.37	0.37	0.23	0.23
BH	N/A	2019-12-31_2022-12-31	N/A	1	No	N/A	0.25	0.25	0.32	0.32
RBC	5y	2019-12-31_2022-12-31	20b	1	No	MOM	0.22	0.22	0.27	0.27
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MOM	0.22	0.22	0.27	0.27
RBC	20y	2019-12-31_2022-12-31	20b	1	No	MOM	0.22	0.22	0.27	0.27
RBC	5y	2019-12-31_2022-12-31	10b	1	No	MOM	0.21	0.21	0.26	0.26
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MOM	0.21	0.21	0.26	0.26
RBC	20y	2019-12-31_2022-12-31	10b	1	No	MOM	0.21	0.21	0.26	0.26
RBC	5y	2019-12-31_2022-12-31	30b	1	No	MOM	0.19	0.19	0.23	0.23
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MOM	0.19	0.19	0.23	0.23
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MOM	0.19	0.19	0.23	0.23
RBC	5y	2019-12-31_2022-12-31	30b	1	No	MR	-0.19	-0.19	-0.23	-0.23
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.19	-0.19	-0.23	-0.23
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MR	-0.19	-0.19	-0.23	-0.23
RBC	5y	2019-12-31_2022-12-31	10b	1	No	MR	-0.21	-0.21	-0.26	-0.26
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MR	-0.21	-0.21	-0.26	-0.26
RBC	20y	2019-12-31_2022-12-31	10b	1	No	MR	-0.21	-0.21	-0.26	-0.26
RBC	5y	2019-12-31_2022-12-31	20b	1	No	MR	-0.22	-0.22	-0.27	-0.27
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MR	-0.22	-0.22	-0.27	-0.27

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	20y	2019-12-31_2022-12-31	20b	1	No	MR	-0.22	-0.22	-0.27	-0.27
RBC	5y	2019-12-31_2022-12-31	30b	2	No	MR	-0.37	-0.37	-0.23	-0.23
RBC	10y	2019-12-31_2022-12-31	30b	2	No	MR	-0.37	-0.37	-0.23	-0.23
RBC	20y	2019-12-31_2022-12-31	30b	2	No	MR	-0.37	-0.37	-0.23	-0.23
RBC	5y	2019-12-31_2022-12-31	10b	2	No	MR	-0.41	-0.41	-0.26	-0.26
RBC	10y	2019-12-31_2022-12-31	10b	2	No	MR	-0.41	-0.41	-0.26	-0.26
RBC	20y	2019-12-31_2022-12-31	10b	2	No	MR	-0.41	-0.41	-0.26	-0.26
RBC	5y	2019-12-31_2022-12-31	20b	2	No	MR	-0.44	-0.44	-0.27	-0.27
RBC	10y	2019-12-31_2022-12-31	20b	2	No	MR	-0.44	-0.44	-0.27	-0.27
RBC	20y	2019-12-31_2022-12-31	20b	2	No	MR	-0.44	-0.44	-0.27	-0.27
RBC	5y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.59	-0.59	-0.49	-0.49
RBC	10y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.59	-0.59	-0.49	-0.49
RBC	20y	2019-12-31_2022-12-31	30b	1	Yes	MR	-0.59	-0.59	-0.49	-0.49
RBC	5y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.59	-0.59	-0.48	-0.48
RBC	10y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.59	-0.59	-0.48	-0.48
RBC	20y	2019-12-31_2022-12-31	10b	1	Yes	MR	-0.59	-0.59	-0.48	-0.48
RBC	5y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.6	-0.6	-0.5	-0.5
RBC	10y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.6	-0.6	-0.5	-0.5
RBC	20y	2019-12-31_2022-12-31	20b	1	Yes	MR	-0.6	-0.6	-0.5	-0.5
RBC	5y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.18	-1.18	-0.49	-0.49
RBC	10y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.18	-1.18	-0.49	-0.49
RBC	20y	2019-12-31_2022-12-31	30b	2	Yes	MR	-1.18	-1.18	-0.49	-0.49
RBC	5y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.19	-1.19	-0.48	-0.48
RBC	10y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.19	-1.19	-0.48	-0.48
RBC	20y	2019-12-31_2022-12-31	10b	2	Yes	MR	-1.19	-1.19	-0.48	-0.48
RBC	5y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.21	-1.2	-0.5	-0.5
RBC	10y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.21	-1.2	-0.5	-0.5
RBC	20y	2019-12-31_2022-12-31	20b	2	Yes	MR	-1.21	-1.2	-0.5	-0.5

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Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.8	0.8	0.69	0.69
RBC	20y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.8	0.8	0.69	0.69
RBC	10y	2022-12-31_2025-09-30	30b	2	No	MR	0.64	0.64	0.76	0.76
RBC	20y	2022-12-31_2025-09-30	30b	2	No	MR	0.64	0.64	0.76	0.76

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
BH	N/A	2022-12-31_2025-09-30	N/A	1	No	N/A	0.62	0.62	1.37	1.37
RBC	10y	2022-12-31_2025-09-30	20b	2	Yes	MR	0.56	0.56	0.48	0.48
RBC	20y	2022-12-31_2025-09-30	20b	2	Yes	MR	0.56	0.56	0.48	0.48
RBC	10y	2022-12-31_2025-09-30	20b	2	No	MR	0.54	0.54	0.64	0.64
RBC	20y	2022-12-31_2025-09-30	20b	2	No	MR	0.54	0.54	0.64	0.64
RBC	5y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.41	0.42	0.27	0.28
RBC	10y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.4	0.4	0.69	0.69
RBC	20y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.4	0.4	0.69	0.69
RBC	5y	2022-12-31_2025-09-30	30b	2	No	MR	0.36	0.36	0.43	0.43
RBC	10y	2022-12-31_2025-09-30	10b	2	Yes	MR	0.34	0.34	0.28	0.28
RBC	20y	2022-12-31_2025-09-30	10b	2	Yes	MR	0.34	0.34	0.28	0.28
RBC	10y	2022-12-31_2025-09-30	10b	2	No	MR	0.33	0.33	0.38	0.38
RBC	20y	2022-12-31_2025-09-30	10b	2	No	MR	0.33	0.33	0.38	0.38
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	10y	2022-12-31_2025-09-30	20b	1	Yes	MR	0.28	0.28	0.48	0.48
RBC	20y	2022-12-31_2025-09-30	20b	1	Yes	MR	0.28	0.28	0.48	0.48
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MR	0.27	0.27	0.64	0.64
RBC	20y	2022-12-31_2025-09-30	20b	1	No	MR	0.27	0.27	0.64	0.64
RBC	5y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.21	0.21	0.27	0.28
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MR	0.18	0.18	0.43	0.43
RBC	10y	2022-12-31_2025-09-30	10b	1	Yes	MR	0.17	0.17	0.28	0.28
RBC	20y	2022-12-31_2025-09-30	10b	1	Yes	MR	0.17	0.17	0.28	0.28
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MR	0.16	0.16	0.38	0.38
RBC	20y	2022-12-31_2025-09-30	10b	1	No	MR	0.16	0.16	0.38	0.38
RBC	5y	2022-12-31_2025-09-30	10b	2	Yes	MOM	0.1	0.1	0.06	0.06
RBC	5y	2022-12-31_2025-09-30	20b	2	No	MR	0.05	0.05	0.06	0.06
RBC	5y	2022-12-31_2025-09-30	10b	1	Yes	MOM	0.05	0.05	0.06	0.06
RBC	5y	2022-12-31_2025-09-30	10b	2	No	MOM	0.04	0.04	0.05	0.05
RBC	5y	2022-12-31_2025-09-30	20b	1	No	MR	0.03	0.03	0.06	0.06
RBC	5y	2022-12-31_2025-09-30	10b	1	No	MOM	0.02	0.02	0.05	0.05
RBC	5y	2022-12-31_2025-09-30	20b	2	Yes	MR	0	0	0	0
RBC	5y	2022-12-31_2025-09-30	20b	1	Yes	MR	0	0	0	0
RBC	5y	2022-12-31_2025-09-30	20b	1	Yes	MOM	0	0	0	0
RBC	5y	2022-12-31_2025-09-30	20b	2	Yes	MOM	0	0	0	0
RBC	5y	2022-12-31_2025-09-30	10b	1	No	MR	-0.02	-0.02	-0.05	-0.05
RBC	5y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.03	-0.03	-0.06	-0.06

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	2022-12-31_2025-09-30	10b	2	No	MR	-0.04	-0.04	-0.05	-0.05
RBC	5y	2022-12-31_2025-09-30	10b	1	Yes	MR	-0.05	-0.05	-0.07	-0.06
RBC	5y	2022-12-31_2025-09-30	20b	2	No	MOM	-0.05	-0.05	-0.06	-0.06
RBC	5y	2022-12-31_2025-09-30	10b	2	Yes	MR	-0.1	-0.1	-0.07	-0.06
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MOM	-0.16	-0.16	-0.38	-0.38
RBC	20y	2022-12-31_2025-09-30	10b	1	No	MOM	-0.16	-0.16	-0.38	-0.38
RBC	10y	2022-12-31_2025-09-30	10b	1	Yes	MOM	-0.17	-0.17	-0.29	-0.28
RBC	20y	2022-12-31_2025-09-30	10b	1	Yes	MOM	-0.17	-0.17	-0.29	-0.28
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.18	-0.18	-0.43	-0.43
RBC	5y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.21	-0.21	-0.28	-0.28
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.27	-0.27	-0.64	-0.64
RBC	20y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.27	-0.27	-0.64	-0.64
RBC	10y	2022-12-31_2025-09-30	20b	1	Yes	MOM	-0.28	-0.28	-0.48	-0.48
RBC	20y	2022-12-31_2025-09-30	20b	1	Yes	MOM	-0.28	-0.28	-0.48	-0.48
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.32	-0.32	-0.76	-0.76
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.32	-0.32	-0.76	-0.76
RBC	10y	2022-12-31_2025-09-30	10b	2	No	MOM	-0.33	-0.33	-0.38	-0.38
RBC	20y	2022-12-31_2025-09-30	10b	2	No	MOM	-0.33	-0.33	-0.38	-0.38
RBC	10y	2022-12-31_2025-09-30	10b	2	Yes	MOM	-0.34	-0.34	-0.29	-0.28
RBC	20y	2022-12-31_2025-09-30	10b	2	Yes	MOM	-0.34	-0.34	-0.29	-0.28
RBC	5y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.36	-0.36	-0.43	-0.43
RBC	10y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.4	-0.4	-0.69	-0.69
RBC	20y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.4	-0.4	-0.69	-0.69
RBC	5y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.42	-0.42	-0.28	-0.28
RBC	10y	2022-12-31_2025-09-30	20b	2	No	MOM	-0.55	-0.54	-0.64	-0.64
RBC	20y	2022-12-31_2025-09-30	20b	2	No	MOM	-0.55	-0.54	-0.64	-0.64
RBC	10y	2022-12-31_2025-09-30	20b	2	Yes	MOM	-0.56	-0.56	-0.48	-0.48
RBC	20y	2022-12-31_2025-09-30	20b	2	Yes	MOM	-0.56	-0.56	-0.48	-0.48
RBC	10y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.64	-0.64	-0.76	-0.76
RBC	20y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.64	-0.64	-0.76	-0.76
RBC	10y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.8	-0.8	-0.69	-0.69
RBC	20y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.8	-0.8	-0.69	-0.69

Dow Jones Industrial Average 12/2022-12/2025

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.88	0.89	0.86	0.86
RBC	20y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.88	0.89	0.86	0.86
RBC	10y	2022-12-31_2025-09-30	30b	2	No	MR	0.71	0.71	0.95	0.95
RBC	20y	2022-12-31_2025-09-30	30b	2	No	MR	0.71	0.71	0.95	0.95
RBC	10y	2022-12-31_2025-09-30	20b	2	Yes	MR	0.5	0.5	0.47	0.47
RBC	20y	2022-12-31_2025-09-30	20b	2	Yes	MR	0.5	0.5	0.47	0.47
RBC	10y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.44	0.44	0.86	0.86
RBC	20y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.44	0.44	0.86	0.86
RBC	5y	2022-12-31_2025-09-30	30b	2	Yes	MR	0.43	0.44	0.33	0.33
RBC	10y	2022-12-31_2025-09-30	20b	2	No	MR	0.43	0.43	0.57	0.58
RBC	20y	2022-12-31_2025-09-30	20b	2	No	MR	0.43	0.43	0.57	0.58
BH	N/A	2022-12-31_2025-09-30	N/A	1	No	N/A	0.4	0.4	1	1
RBC	5y	2022-12-31_2025-09-30	30b	2	No	MR	0.37	0.37	0.5	0.5
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.36	0.36	0.95	0.95
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MR	0.36	0.36	0.95	0.95
RBC	10y	2022-12-31_2025-09-30	10b	2	Yes	MR	0.29	0.29	0.26	0.26
RBC	20y	2022-12-31_2025-09-30	10b	2	Yes	MR	0.29	0.29	0.26	0.26
RBC	10y	2022-12-31_2025-09-30	20b	1	Yes	MR	0.25	0.25	0.47	0.47
RBC	20y	2022-12-31_2025-09-30	20b	1	Yes	MR	0.25	0.25	0.47	0.47
RBC	10y	2022-12-31_2025-09-30	10b	2	No	MR	0.24	0.24	0.31	0.32
RBC	20y	2022-12-31_2025-09-30	10b	2	No	MR	0.24	0.24	0.31	0.32
RBC	5y	2022-12-31_2025-09-30	30b	1	Yes	MR	0.22	0.22	0.33	0.33
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MR	0.22	0.22	0.57	0.58
RBC	20y	2022-12-31_2025-09-30	20b	1	No	MR	0.22	0.22	0.57	0.58
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MR	0.19	0.19	0.5	0.5
RBC	10y	2022-12-31_2025-09-30	10b	1	Yes	MR	0.14	0.14	0.26	0.26
RBC	20y	2022-12-31_2025-09-30	10b	1	Yes	MR	0.14	0.14	0.26	0.26
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MR	0.12	0.12	0.31	0.32
RBC	20y	2022-12-31_2025-09-30	10b	1	No	MR	0.12	0.12	0.31	0.32
RBC	5y	2022-12-31_2025-09-30	20b	2	Yes	MOM	0.09	0.1	0.07	0.07
RBC	5y	2022-12-31_2025-09-30	20b	2	No	MOM	0.08	0.08	0.11	0.11
RBC	5y	2022-12-31_2025-09-30	10b	2	Yes	MOM	0.06	0.06	0.05	0.05
RBC	5y	2022-12-31_2025-09-30	20b	1	Yes	MOM	0.05	0.05	0.07	0.07
RBC	5y	2022-12-31_2025-09-30	20b	1	No	MOM	0.04	0.04	0.11	0.11
RBC	5y	2022-12-31_2025-09-30	10b	1	Yes	MOM	0.03	0.03	0.05	0.05
RBC	5y	2022-12-31_2025-09-30	10b	2	No	MOM	0.02	0.02	0.02	0.02
RBC	5y	2022-12-31_2025-09-30	10b	1	No	MOM	0.01	0.01	0.02	0.02

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	2022-12-31_2025-09-30	10b	1	No	MR	-0.01	-0.01	-0.02	-0.02
RBC	5y	2022-12-31_2025-09-30	10b	2	No	MR	-0.02	-0.02	-0.02	-0.02
RBC	5y	2022-12-31_2025-09-30	10b	1	Yes	MR	-0.03	-0.03	-0.05	-0.05
RBC	5y	2022-12-31_2025-09-30	20b	1	No	MR	-0.04	-0.04	-0.11	-0.11
RBC	5y	2022-12-31_2025-09-30	20b	1	Yes	MR	-0.05	-0.05	-0.07	-0.07
RBC	5y	2022-12-31_2025-09-30	10b	2	Yes	MR	-0.06	-0.06	-0.05	-0.05
RBC	5y	2022-12-31_2025-09-30	20b	2	No	MR	-0.09	-0.08	-0.11	-0.11
RBC	5y	2022-12-31_2025-09-30	20b	2	Yes	MR	-0.1	-0.1	-0.07	-0.07
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MOM	-0.12	-0.12	-0.32	-0.32
RBC	20y	2022-12-31_2025-09-30	10b	1	No	MOM	-0.12	-0.12	-0.32	-0.32
RBC	10y	2022-12-31_2025-09-30	10b	1	Yes	MOM	-0.15	-0.14	-0.27	-0.26
RBC	20y	2022-12-31_2025-09-30	10b	1	Yes	MOM	-0.15	-0.14	-0.27	-0.26
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.19	-0.19	-0.5	-0.5
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.22	-0.22	-0.58	-0.58
RBC	20y	2022-12-31_2025-09-30	20b	1	No	MOM	-0.22	-0.22	-0.58	-0.58
RBC	5y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.22	-0.22	-0.33	-0.33
RBC	10y	2022-12-31_2025-09-30	10b	2	No	MOM	-0.24	-0.24	-0.32	-0.32
RBC	20y	2022-12-31_2025-09-30	10b	2	No	MOM	-0.24	-0.24	-0.32	-0.32
RBC	10y	2022-12-31_2025-09-30	20b	1	Yes	MOM	-0.25	-0.25	-0.47	-0.47
RBC	20y	2022-12-31_2025-09-30	20b	1	Yes	MOM	-0.25	-0.25	-0.47	-0.47
RBC	10y	2022-12-31_2025-09-30	10b	2	Yes	MOM	-0.29	-0.29	-0.27	-0.26
RBC	20y	2022-12-31_2025-09-30	10b	2	Yes	MOM	-0.29	-0.29	-0.27	-0.26
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.36	-0.36	-0.96	-0.95
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MOM	-0.36	-0.36	-0.96	-0.95
RBC	5y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.37	-0.37	-0.5	-0.5
RBC	10y	2022-12-31_2025-09-30	20b	2	No	MOM	-0.43	-0.43	-0.58	-0.58
RBC	20y	2022-12-31_2025-09-30	20b	2	No	MOM	-0.43	-0.43	-0.58	-0.58
RBC	5y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.44	-0.44	-0.33	-0.33
RBC	10y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.44	-0.44	-0.86	-0.86
RBC	20y	2022-12-31_2025-09-30	30b	1	Yes	MOM	-0.44	-0.44	-0.86	-0.86
RBC	10y	2022-12-31_2025-09-30	20b	2	Yes	MOM	-0.5	-0.5	-0.47	-0.47
RBC	20y	2022-12-31_2025-09-30	20b	2	Yes	MOM	-0.5	-0.5	-0.47	-0.47
RBC	10y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.72	-0.71	-0.96	-0.95
RBC	20y	2022-12-31_2025-09-30	30b	2	No	MOM	-0.72	-0.71	-0.96	-0.95
RBC	10y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.89	-0.89	-0.86	-0.86
RBC	20y	2022-12-31_2025-09-30	30b	2	Yes	MOM	-0.89	-0.89	-0.86	-0.86

Results show that mean-reversion strategy variants outperform momentum ones throughout most of the horizon, except for 12/2007-12/2009 and 12/2019-12/2022 time periods. Longer data lengths and signal lags tend to improve performance. Similarly, leverage and doubling down also increase PnL, although have little impact on SR. These features are generally required to beat the buy-and-hold strategy. S&P 500 and DJIA assets produce comparable outcomes. Results are explored in more detail throughout 5. RESULTS.

Appendix E: Extended Results: Strategy Results across Data Lengths

S&P500 30b Signal Lag All Horizons

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	30b	1	No	MR	1.16	1.16	0.23	0.23
RBC	5y	1999-12-31_2025-09-30	30b	1	No	MR	0.98	0.98	0.2	0.2
RBC	5y	1999-12-31_2007-12-31	30b	1	No	MR	0.56	0.57	0.39	0.39
RBC	10y	1999-12-31_2007-12-31	30b	1	No	MR	0.27	0.27	0.19	0.19
RBC	10y	2007-12-31_2009-12-31	30b	1	No	MR	-0.5	-0.5	-0.67	-0.67
RBC	5y	2007-12-31_2009-12-31	30b	1	No	MR	-0.56	-0.56	-0.76	-0.76
RBC	10y	2009-12-31_2019-12-31	30b	1	No	MR	1.4	1.4	0.81	0.81
RBC	20y	2009-12-31_2019-12-31	30b	1	No	MR	1.25	1.25	0.72	0.72
RBC	5y	2009-12-31_2019-12-31	30b	1	No	MR	1.18	1.18	0.68	0.68
RBC	5y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	10y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	20y	2019-12-31_2022-12-31	30b	1	No	MR	-0.2	-0.2	-0.25	-0.25
RBC	10y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	20y	2022-12-31_2025-09-30	30b	1	No	MR	0.32	0.32	0.76	0.76
RBC	5y	2022-12-31_2025-09-30	30b	1	No	MR	0.18	0.18	0.43	0.43

S&P500 20b Signal Lag All Horizons

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	20b	1	No	MR	0.92	0.92	0.18	0.18
RBC	5y	1999-12-31_2025-09-30	20b	1	No	MR	0.64	0.64	0.13	0.13
RBC	5y	1999-12-31_2007-12-31	20b	1	No	MR	0.48	0.48	0.33	0.33
RBC	10y	1999-12-31_2007-12-31	20b	1	No	MR	0.16	0.16	0.11	0.11
RBC	10y	2007-12-31_2009-12-31	20b	1	No	MR	-0.43	-0.43	-0.58	-0.58

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	5y	2007-12-31_2009-12-31	20b	1	No	MR	-0.5	-0.5	-0.68	-0.68
RBC	10y	2009-12-31_2019-12-31	20b	1	No	MR	1.3	1.3	0.75	0.75
RBC	20y	2009-12-31_2019-12-31	20b	1	No	MR	1.28	1.28	0.74	0.74
RBC	5y	2009-12-31_2019-12-31	20b	1	No	MR	1.05	1.05	0.6	0.6
RBC	5y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	10y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	20y	2019-12-31_2022-12-31	20b	1	No	MR	-0.29	-0.29	-0.35	-0.35
RBC	10y	2022-12-31_2025-09-30	20b	1	No	MR	0.27	0.27	0.64	0.64
RBC	20y	2022-12-31_2025-09-30	20b	1	No	MR	0.27	0.27	0.64	0.64
RBC	5y	2022-12-31_2025-09-30	20b	1	No	MR	0.03	0.03	0.06	0.06

S&P500 10b Signal Lag All Horizons

Model	Data Length	Horizon	Signal Lag	Leverage	Double Down	Strategy Type	Net PnL	Gross PnL	Net SR	Gross SR
RBC	10y	1999-12-31_2025-09-30	10b	1	No	MR	0.51	0.51	0.1	0.1
RBC	5y	1999-12-31_2025-09-30	10b	1	No	MR	0.37	0.37	0.07	0.07
RBC	5y	1999-12-31_2007-12-31	10b	1	No	MR	0.26	0.26	0.18	0.18
RBC	10y	1999-12-31_2007-12-31	10b	1	No	MR	-0.16	-0.16	-0.11	-0.11
RBC	5y	2007-12-31_2009-12-31	10b	1	No	MR	-0.47	-0.47	-0.64	-0.64
RBC	10y	2007-12-31_2009-12-31	10b	1	No	MR	-0.54	-0.54	-0.72	-0.72
RBC	20y	2009-12-31_2019-12-31	10b	1	No	MR	1.43	1.43	0.82	0.82
RBC	10y	2009-12-31_2019-12-31	10b	1	No	MR	1.42	1.42	0.82	0.82
RBC	5y	2009-12-31_2019-12-31	10b	1	No	MR	1.06	1.06	0.61	0.61
RBC	5y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	10y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	20y	2019-12-31_2022-12-31	10b	1	No	MR	-0.26	-0.26	-0.32	-0.32
RBC	10y	2022-12-31_2025-09-30	10b	1	No	MR	0.16	0.16	0.38	0.38
RBC	20y	2022-12-31_2025-09-30	10b	1	No	MR	0.16	0.16	0.38	0.38
RBC	5y	2022-12-31_2025-09-30	10b	1	No	MR	-0.02	-0.02	-0.05	-0.05

Strategy variants with 10 years data length continue to be the most successful variants across varying signal lags. This further reinforces the findings in 5.4. Data Length and supports the choice of the 10y setting in optimal strategy setup.

Appendix F: Strategy Labelling Convention

The labelling of strategies throughout 5. RESULTS follows the convention below.

Strategy labelling: M_DL_H_SL_L_DD_SC

Variable	Values	Meaning
M	*rbc*	Model
DL	*5cal* *10cal* *20cal*	Data length (years for model training)
H	*1999-12-31_2025-09-30* *1999-12-31_2007-12-31* *2007-12-31_2009-12-31* *2009-12-31_2019-12-31* *2019-12-31_2022-12-31* *2022-12-31_2025-09-30*	Horizon (date range)
SL	*10lag* *20lag* *30lag*	Signal lag (business days)
L	*2lev* *none*	Leverage (2x or none)
DD	*dd* *none*	Double down (binary)
S	*mom* *mr*	Strategy type (momentum or mean-reversion)
C	*g* *n*	Gross vs net